Probabilistic Spatiotemporal Layered Video Transmission in Joint Channel Under Strict Rate-Floor & Rate-Ceiling Limits

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Abstract

Video links that deliver a few megabits per second must choose between a conventional pipeline, whose bit rate is fixed before the channel is observed and whose quality collapses when the capacity falls below that rate, and a linear analogue-style pipeline, which degrades gracefully but offers no stated guarantee. We study a two-layer hybrid digital–analogue transmission with a probabilistic quality floor. We define a rate–distortion-equivalent throughput on a state-independent reference scale, a chance-constrained floor of two megabits per second equivalent, and a pre-channel distortion fraction. We prove a converse bound on the equivalent throughput; a closed-form price for isolating the floor from error propagation; a floor-isolation theorem; closed-form power allocations for the analogue layer with receiver side information, channel-estimation error, and a peak-power limit; a certified finite-blocklength guarantee for the floor block, whose dispersion penalty is below two percent of the channel uses at the target scale so that fading diversity governs feasibility; and a margin rule for delayed capacity feedback. A pilot on one previously compressed 640×352 clip shows that the floor-only reconstruction never falls below 31.6 dB, that the analytic model predicts the measured quality within 0.4 dB, and that the hybrid design improves on a linear analogue design by 8 to 10 dB at a bandwidth ratio of 0.17. It also shows that the design remains 3.3 to 3.5 dB below zero-latency H.264 followed by an ideal capacity-achieving code, at a floor block of 1.9 times the credited payload. We state the regime in which a digital codec should still be preferred and the experiments required to test the remaining claims.

Keywords: joint source–channel coding; hybrid digital–analogue transmission; finite-blocklength information theory; chance-constrained design; Wyner–Ziv side information; wireless video.

1 Introduction

1.1 Operating Regime and Motivation

Consider progressive video with ρHV samples per frame, where $H \times V$ is the pixel grid and ρ is the number of samples per pixel ($\rho = 3/2$ for 4:2:0 chroma subsampling), a bit depth of β bits, and a frame rate of f_v frames per second. The raw rate is

$$R_{\text{raw}} = \beta\, \rho\, HV\, f_v. \tag{1}$$

A delivered rate of T megabits per second (Mbps) corresponds to a reduction ratio $\kappa = R_{\text{raw}}/T$ and to $T/(\rho HV f_v)$ bits per sample. Table 1 evaluates these quantities for three common formats, and Table 2 gives the Shannon capacity of an additive white Gaussian noise channel for representative bandwidths and signal-to-noise ratios. Two facts follow directly. First, at a delivered rate of 2 to 10 Mbps the information content per sample is far below one bit, so any scheme must remove most of the raw information; what remains open is *where* and *how* that removal occurs. Second, a 2 to 10 Mbps requirement is of the order of the capacity of a narrowband link at moderate signal-to-noise ratio (first rows of Table 2), so the floor cannot be met by operating near capacity at the lowest bandwidths and signal-to-noise ratios, and the architecture must specify how much margin it retains.

Table 1: Raw rate of progressive 8-bit 4:2:0 video at 30 frames per second, the reduction ratio $\kappa = R_{\text{raw}}/T$ implied by a delivered rate T of 2 and 10 megabits per second, the information content per sample, and the complex symbol rate $\rho HV f_v/2$ needed to send every sample uncoded (two real samples per complex channel use).

Format	Samples per frame	R_{raw} [Mbps]	κ at 2 Mbps	κ at 10 Mbps	bit/sample (2 / 10 Mbps)	Uncoded symbol rate [Msym/s]
640×480	460,800	110.592	55.30	11.06	0.145 / 0.723	6.912
1280×720	1,382,400	331.776	165.89	33.18	0.048 / 0.241	20.736
1920×1080	3,110,400	746.496	373.25	74.65	0.021 / 0.107	46.656

Table 2: Shannon capacity $W \log_2(1+\gamma)$ in megabits per second of an additive white Gaussian noise channel with W complex symbols per second and signal-to-noise ratio γ, ideal Nyquist signalling, no protocol overhead.

W [MHz]	$\gamma = 5$ dB	$\gamma = 10$ dB	$\gamma = 20$ dB
1	2.06	3.46	6.66
5	10.29	17.30	33.29
20	41.15	69.19	133.16

1.2 Limitations of Channel-Blind Upfront Compression

In a conventional pipeline the video encoder, which combines motion-compensated prediction, transform coding, and entropy coding [1, 2], emits a bit stream whose rate is fixed from an estimate of channel quality, and a channel code then protects that stream. Three consequences matter here.

(i) *Threshold behaviour.* Once the instantaneous capacity falls below the code rate, decoding fails abruptly; this is the cliff effect documented for separation-based video and image transmission [3, 4]. When capacity exceeds the code rate, the surplus is unused.

(ii) *Inter-frame dependence.* Prediction loops make a decoding failure in one frame propagate to later frames until a refresh.

(iii) *Encoder-side burden.* The reduction ratios of Table 1 are achieved at the encoder, before any channel information is available at that granularity.

The source–channel separation theorem [5, 6] guarantees that this decomposition loses nothing in the asymptotic limit for a known channel. It does not do so at finite blocklength [7], nor when the source and channel are not matched in the sense of [8], nor when the channel varies faster than the rate can be updated. Each of these conditions is present in the target regime.

1.3 Problem Statement

Problem 1 (Informal statement)**.** Design an encoder–decoder pair for video transmission over a time-varying wireless channel such that (a) the quality delivered to the receiver, expressed as a rate–distortion-equivalent throughput, is at least 2 Mbps with probability at least $1 - \epsilon$ in every group of pictures; (b) that quantity tracks the instantaneous channel capacity, up to 10 Mbps, when the channel permits; (c) the encoder performs as little channel-blind, irreversible compression as possible; and (d) latency, peak-power, and encoder-complexity budgets are respected.

1.4 Adjacent Challenges

?? 1 is coupled to a set of difficulties that any solution must confront jointly. Table 3 lists them with their mathematical form and their specific weight at the target operating point.

Table 3: Challenges adjacent to the problem and their mathematical form. Here γ is the signal-to-noise ratio, $k = GN$ the source dimension per group of pictures, n the number of complex channel uses, V the channel dispersion, and Q^{-1} the inverse Gaussian complementary cumulative distribution function.

Challenge	Mathematical form	Weight at 2–10 Mbps
Threshold (cliff) behaviour	End-to-end distortion $D(\gamma)$ is discontinuous at the design value γ_0 for a fixed-rate digital chain	A floor requires worst-case reasoning over the capacity distribution
Bandwidth mismatch	Bandwidth ratio $b = 2n/k \neq 1$; symbol-by-symbol (uncoded) mapping is optimal only under probabilistic matching [8]	$b < 1$ in the first two rows of Table 1 unless $W \geq 20.7$ MHz
Non-Gaussian, non-stationary source	Coefficient variances σ_i^2 vary over space and time; reverse water-filling needs them at the receiver	Side-information overhead competes with a payload below 0.25 bit/sample
Temporal redundancy without an encoder prediction loop	Wyner–Ziv function $\geq$ conditional rate–distortion function [9, 10]	Removing motion search moves the burden to the decoder
Peak power and amplifier nonlinearity	Peak-to-average power ratio of continuous-amplitude symbols in multicarrier modulation	Clipping distortion is not corrected by a channel code
Channel-estimation error	Mismatch between estimated and true gains h_j perturbs the linear estimator	Dominant on mobile and aerial links
Short blocklength	Rate penalty $\sqrt{V/n}\,Q^{-1}(\epsilon)$ [11]; joint penalty [7]	Low latency forces small n per group of pictures
Erasures and bursts	Erasure patterns violate the additive-Gaussian-noise model	Spreading transforms trade robustness against uniformity of loss
Distortion measure	Mean-squared error is the analytic measure; perceptual quality differs	$\mathcal{T}_{\text{eq}}$ is defined for squared error and requires a validity argument
Rate adaptation under feedback delay	Control of the floor/refinement split from a delayed estimate of the capacity process	Stability and regret of the adaptation rule

1.5 Contributions

(a) A quality scale that places digital and analogue schemes on one axis, the reference rate–distortion-equivalent throughput $\mathcal{T}_{\text{eq}}$; a chance-constrained floor–ceiling specification in that scale; and a dimensionless measure of channel-blind compression, the pre-channel distortion fraction (Sections 3 and 4).

(b) A converse in surplus form, a necessary feasibility condition, a closed-form price of isolating the floor from error propagation, and a floor-isolation theorem (Sections 3 to 5).

(c) Closed-form allocations for the analogue layer with receiver side information, with channel-estimation error, and with a hard amplitude limiter; a first-order optimality condition for the floor power and the channel-use split; and a multiplicative robustness bound (Sections B, C and 5).

(d) A certified finite-blocklength guarantee for the floor block, the distinction between the dispersions of i.i.d. and equal-energy codebooks, the diversity requirement, a chance-constrained

margin rule for delayed feedback, and the separation of safety from performance (Sections C and 6).

(e) A reproducible reference implementation and a pilot evaluation, including results unfavourable to the design (Sections D and 7).

1.6 Organisation

Section 2 positions the problem; Section 3 formalises it; Sections 4 and 5 give the model and the architecture; Section 6 certifies the floor block; Sections B and C treat resource allocation, peak power, estimation, feedback and control; Section D reports the evaluation protocol and pilot; and Sections 7 to 9 discuss what the evidence supports, the limitations, the remaining work, and the conclusion.

2 Related Work and Gap Analysis

Separation and its limits. Shannon's separation result [5,6,12] underlies the standard design. Gastpar, Rimoldi and Vetterli showed that optimality requires the source and the channel to be matched in a probabilistic sense, a condition that contains separation and uncoded transmission as special cases [8]. At finite blocklength, Kostina and Verdú established that joint source–channel code design has a measurable advantage over separate design, through a dispersion-based characterisation $nC - kR(d) \approx \sqrt{nV + k\mathcal{V}(d)}\, Q^{-1}(\epsilon)$ [7], building on the channel-coding analysis of [11].

Standard codecs and layered coding. Hybrid codecs [1,2] deliver the compression ratios of Table 1 by prediction and entropy coding. Scalable extensions [13] provide quality layers but retain hard-decision layers and prediction dependencies.

Distributed video coding. Building on the Slepian–Wolf and Wyner–Ziv results [9, 14], distributed video coding moves prediction to the decoder and thereby lowers encoder complexity [10]. It addresses encoder burden but not, by itself, channel adaptation.

Linear and analogue-style video transmission. SoftCast makes the transmitted samples linearly related to pixel luminance, so that channel noise degrades sharpness rather than causing glitches, and reports average multicast quality gains of 5.5 dB over a conventional design in its testbed [15,16].

Hybrid digital–analogue coding and successive refinement. Hybrid digital–analogue schemes combine the two regimes [17], and successive refinement formalises layered description [18]. These results are theoretical or Gaussian-source results rather than complete video systems.

Learned joint source–channel coding. Deep joint source–channel coding maps images directly to channel symbols and avoids the cliff effect [4]; it has been extended to bandwidth-agile successive refinement [19] and to video groups of pictures [3], which reports gains over H.264 and H.265 followed by low-density parity-check coding in structural similarity. The guarantees are empirical, and the encoder is a trained neural network.

Rateless coding. Fountain-type codes [20,21] adapt redundancy to the channel without a fixed rate and are natural candidates for a floor layer.

Table 4 summarises a qualitative comparison. The assessment is the authors' reading of the cited works, restricted to the literature surveyed here.

Table 4: Qualitative comparison of transmission approaches against the requirements of **??** 1. "Stated" means the cited work states the property; "none stated" means the surveyed work does not state it.

Approach	Encoder-side reduction	Graceful degradation	Throughput floor guarantee	Receiver-side side information
Hybrid codec plus channel code [1, 2]	Heavy, channel-blind	Cliff effect [3]	Only by conservative rate choice	No
Scalable video coding [13]	Heavy, layered	Stepwise	Base layer	No
Distributed video coding [10]	Light	Not addressed	None stated	Yes (decoder-side prediction)
Linear mapping [16]	Light, transform only	Yes	None stated	Not in the cited design
Learned joint coding [3, 19]	Neural network	Yes	Empirical	Learned
Hybrid digital–analogue [17]	Scheme-dependent	Yes (theory)	None stated	Not in the cited design

Gap. In the surveyed works we find no formulation that jointly states (i) a probabilistic floor on delivered quality in a scale common to digital and analogue layers, (ii) a quantitative measure of channel-blind irreversible compression, and (iii) a design whose floor layer is calibrated by finite-blocklength dispersion while its refinement layer exploits receiver-side side information. Section 3 supplies (i) and (ii); the later sections target (iii).

3 Formal Problem Statement

3.1 Source, Channel, and Codes

Source. A group of pictures (GoP) of G frames, with $N = \rho HV$ samples per frame, is the vector $\boldsymbol{X}_g \in \mathbb{R}^k$ with $k = GN$. Its duration is $T_g = G/f_v$ seconds. The distortion between $\boldsymbol{x}$ and $\hat{\boldsymbol{x}}$ is $d(\boldsymbol{x}, \hat{\boldsymbol{x}}) = k^{-1}\|\boldsymbol{x} - \hat{\boldsymbol{x}}\|^2$.

Receiver-side information. $\boldsymbol{Z}_g$ denotes information available only at the receiver before decoding group g, for example previously reconstructed GoPs. Conditionally on the past, $\boldsymbol{Z}_g$ is independent of the noise of the current GoP. The case $\boldsymbol{Z}_g = \varnothing$ is allowed.

Channel. Group g is carried by $n = \lfloor (1-\varphi)\, W\, T_g \rfloor$ complex channel uses, where W is the complex symbol rate and $\varphi \in [0, 1)$ the fraction of resources spent on pilots and protocol overhead. For $j = 1, \ldots, n$,

$$Y_{g,j} = h_{g,j}\, S_{g,j} + Z^{\mathrm{ch}}_{g,j}, \qquad Z^{\mathrm{ch}}_{g,j} \sim \mathcal{CN}(0, \nu), \tag{2}$$

with channel gains $h_{g,j}$ known to the receiver and independent of the source and of the noise, and with the power constraint $\sum_{j=1}^{n} \mathbb{E}|S_{g,j}|^2 \le n\bar{P}$. The vector h_g is the realisation for group g; $\gamma_{g,j} = |h_{g,j}|^2 \bar{P}/\nu$.

Codes. An encoder is a (possibly randomised) map $E : \mathbb{R}^k \to \mathbb{C}^n$, and a decoder is a map $D : \mathbb{C}^n \times \mathcal{Z} \to \mathbb{R}^k$. The distortion conditional on the channel realisation is $D_g(h) =$

$\mathbb{E}[d(\boldsymbol{X}_g, D(\boldsymbol{Y}_g, \boldsymbol{Z}_g)) \mid h_g = h]$.

3.2 Rate–Distortion-Equivalent Throughput

Definition 1 (Information function with decoder side information)**.** For $D \geq 0$, let

$$\mathcal{R}_g(D) = \inf \Big\{ I(\boldsymbol{X};U) - I(\boldsymbol{Z};U) \;:\; U - \boldsymbol{X} - \boldsymbol{Z},\ \mathbb{E}\, d(\boldsymbol{X}, f(U,\boldsymbol{Z})) \leq D \Big\}, \tag{3}$$

where the infimum is over conditional laws $P_{U|\boldsymbol{X}}$ and reconstruction maps f ($\boldsymbol{Z}$ is taken jointly distributed with $\boldsymbol{X}$ as above). For $\boldsymbol{Z} = \varnothing$ this is the information rate–distortion function $\inf\{I(\boldsymbol{X};\hat{\boldsymbol{X}}) : \mathbb{E}\, d(\boldsymbol{X},\hat{\boldsymbol{X}}) \leq D\}$ [12]. It is non-increasing in D, and we write $\mathcal{D}_g(r) = \inf\{D : \mathcal{R}_g(D) \leq r\}$ for its inverse (the distortion–rate function).

Definition 2 (Rate–distortion-equivalent throughput)**.** For a scheme with conditional distortion $D_g(h)$, the *rate–distortion-equivalent throughput* of group g under realisation h is

$$\mathcal{T}_{\mathrm{eq}}(g,h) = \frac{\mathcal{R}_g(D_g(h))}{T_g} \quad [\text{bit/s}]. \tag{4}$$

Remark 1. $\mathcal{T}_{\mathrm{eq}}$ is the rate at which an optimal source code with the same side information would have to run to reach the distortion the scheme actually achieves. It is defined for any scheme, digital or analogue, and equals the delivered bit rate for an ideal digital scheme that meets its distortion target. An operational variant replaces $\mathcal{R}_g$ by the measured rate of a fixed reference codec at equal distortion; the information function is used here because it is codec-independent. For natural video $\mathcal{R}_g$ is not known in closed form; the Gaussian-surrogate and reference-codec estimates used in the evaluation are specified later and are bounds, not exact values.

3.3 Measuring Upfront Compression

Definition 3 (Pre-channel distortion fraction)**.** Let $\boldsymbol{S} = E(\boldsymbol{X})$ and $D_{\mathrm{enc}} = \inf_{\phi} \mathbb{E}\, d(\boldsymbol{X}, \phi(\boldsymbol{S},\boldsymbol{Z}))$, the distortion left when the *noiseless* encoder output is decoded optimally. Let $D_{\mathrm{tot}} = \mathbb{E}\, d(\boldsymbol{X}, D(\boldsymbol{Y},\boldsymbol{Z}))$ be the end-to-end distortion of the scheme. The *pre-channel distortion fraction* is

$$\beta_{\mathrm{up}} = \frac{D_{\mathrm{enc}}}{D_{\mathrm{tot}}}. \tag{5}$$

Lemma 1. *$0 \leq D_{\mathrm{enc}} \leq D_{\mathrm{tot}}$; hence $\beta_{\mathrm{up}} \in [0,1]$ whenever $D_{\mathrm{tot}} > 0$.*

Proof. Fix the decoder D and define the randomised reconstruction $\phi_\omega(,) = D(\mathrm{channel}_\omega(),)$, in which the channel noise ω is generated at the receiver. Its expected distortion equals D_{tot}. For every , there is a deterministic choice of ω whose conditional expected distortion does not exceed the average over ω; minimising over such choices gives a deterministic ϕ with distortion at most D_{tot}. Hence $D_{\mathrm{enc}} \leq D_{\mathrm{tot}}$. □

Proposition 1 (Fixed-rate digital pre-compression)**.** *If $E(\boldsymbol{X})$ takes at most 2^r distinct values, then $D_{\mathrm{enc}} \geq \mathcal{D}_g(r)$ and therefore*

$$\beta_{\mathrm{up}} \;\geq\; \frac{\mathcal{D}_g(r)}{D_{\mathrm{tot}}}.$$

In particular $\beta_{\mathrm{up}} > 0$ whenever $\mathcal{D}_g(r) > 0$, that is, whenever r is below the rate needed for zero distortion.

Proof. Let $U = \boldsymbol{S}$ and $\hat{\boldsymbol{X}} = \phi(U,\boldsymbol{Z})$ attain D_{enc}. The chain $U - \boldsymbol{X} - \boldsymbol{Z}$ holds because $\boldsymbol{S}$ is a function of $\boldsymbol{X}$ and independent randomness. By definition, $\mathcal{R}_g(D_{\mathrm{enc}}) \leq I(\boldsymbol{X};U) - I(\boldsymbol{Z};U) \leq I(\boldsymbol{X};U) \leq H(U) \leq r$. Since $\mathcal{R}_g$ is non-increasing, $D_{\mathrm{enc}} \geq \mathcal{D}_g(r)$. □

Remark 2. An encoder whose map is injective on the support of $\boldsymbol{X}$, or whose only irreversible step is a discarding of coefficients whose energy is accounted for explicitly, has β_{up} equal to the corresponding discarded fraction. The architecture developed in the later sections is built so that the composite encoder is information-lossless except for an explicit, optimised truncation.

3.4 Converse

Write the capacity of the parallel channels (2) under the power constraint as

$$C_n(h) = \max_{\substack{P_j \geq 0 \\ \sum_j P_j \leq n\bar{P}}} \sum_{j=1}^{n} \log_2\Big(1 + \frac{|h_j|^2 P_j}{\nu}\Big) \quad \text{[bit per GoP]}, \tag{6}$$

which is the water-filling capacity.

Proposition 2 (Converse on rate–distortion-equivalent throughput)**.** *For every encoder–decoder pair and every channel realisation h,*

$$\mathcal{T}_{\text{eq}}(g, h) \;\leq\; \frac{C_n(h)}{T_g}. \tag{7}$$

Proof. Condition on $h_g = h$ and on the past. Let $U = \boldsymbol{Y}^n$ and $\hat{\boldsymbol{X}} = D(U, \boldsymbol{Z})$. Because $\boldsymbol{Z}$ is independent of the channel noise and the encoder does not observe $\boldsymbol{Z}$, the chain $\boldsymbol{Z} - \boldsymbol{X} - U$ holds, so U is feasible in (3) at $D = D_g(h)$. Therefore

$$\begin{aligned} \mathcal{R}_g(D_g(h)) &\leq I(\boldsymbol{X}; U) - I(\boldsymbol{Z}; U) \leq I(\boldsymbol{X}; \boldsymbol{Y}^n) \leq I(\boldsymbol{S}^n; \boldsymbol{Y}^n) \\ &\leq \sum_{j=1}^{n} I(S_j; Y_j) \leq \sum_{j=1}^{n} \log_2\Big(1 + \frac{|h_j|^2 P_j}{\nu}\Big) \leq C_n(h), \end{aligned}$$

where the third inequality is data processing along $\boldsymbol{X} - \boldsymbol{S}^n - \boldsymbol{Y}^n$; the fourth uses $I(\boldsymbol{S}^n; \boldsymbol{Y}^n) = h(\boldsymbol{Y}^n) - h(\boldsymbol{Z}^{\text{ch}})$ with independent noise components and $h(\boldsymbol{Y}^n) \leq \sum_j h(Y_j)$; the fifth uses the maximum-entropy property of the Gaussian with $\mathbb{E}|Y_j|^2 = |h_j|^2 P_j + \nu$, $P_j = \mathbb{E}|S_j|^2$; and the last maximises over allocations $\{P_j\}$ meeting the power constraint. Dividing by T_g gives (7). □

Corollary 1 (Necessary condition for the floor)**.** *If a scheme satisfies $\mathbb{P}\big[\mathcal{T}_{\text{eq}}(g, h) \geq T_{\min}\big] \geq 1 - \epsilon$, then*

$$\mathbb{P}\big[\, C_n(h)/T_g \geq T_{\min} \,\big] \;\geq\; 1 - \epsilon.$$

That is, the ϵ-outage capacity must be at least $T_{\min}$.

Proof. By Proposition 2, the event $\{\mathcal{T}_{\text{eq}} \geq T_{\min}\}$ is contained in $\{C_n(h)/T_g \geq T_{\min}\}$. □

3.5 The Problem

Problem 2 (Probabilistic floor–ceiling transmission)**.** Given $T_{\min} = 2$ Mbps, $T_{\max} = 10$ Mbps, an outage level $\epsilon \in (0, 1)$, an efficiency $\eta \in (0, 1]$, a latency budget $L_{\max}$, a peak-to-average power ratio bound Λ with violation probability δ, and an encoder complexity budget $\kappa_{\max}$ operations per source sample, find (E, D) solving

$$\begin{aligned} \min_{E,D} \quad & \beta_{\text{up}} && (8) \\ \text{s.t.} \quad & \mathbb{P}\big[\mathcal{T}_{\text{eq}}(g, h) \geq T_{\min}\big] \geq 1 - \epsilon, && (\text{F}) \\ & \mathbb{E}\big[\mathcal{T}_{\text{eq}}(g, h)\big] \;\geq\; \mathbb{E}\Big[\min\Big\{T_{\max},\; \eta\,\tfrac{C_n(h)}{T_g}\Big\}\Big], && (\text{R}) \\ & T_g + \tau_{\text{proc}} \leq L_{\max}, && (\text{L}) \\ & \sum_{j=1}^{n} \mathbb{E}|S_j|^2 \leq n\bar{P}, \quad \mathbb{P}\big[\text{PAPR}(\boldsymbol{S}) > \Lambda\big] \leq \delta, && (\text{P}) \\ & \text{encoder operations per sample} \;\leq\; \kappa_{\max}. && (\text{C}) \end{aligned}$$

Remark 3. Proposition 2 gives $\mathcal{T}_{\text{eq}} \le C_n(h)/T_g$, so (R) is meaningful for $\eta < 1$ and is an efficiency requirement relative to the converse. Corollary 1 is a necessary feasibility condition for (F); it is not sufficient, since the guarantee also depends on finite-blocklength loss and on side-information quality, which the floor layer analysis addresses later.

4 Transform, Statistical Model, and Receiver-Side Prediction

4.1 Orthonormal Spatio-Temporal Transform

Definition 4 (Transform)**.** Let $\Psi_{\text{t}} \in \mathbb{R}^{G\times G}$ be an orthonormal temporal transform (Haar or discrete cosine transform of type II [22]) and $\Psi_{\text{s}} \in \mathbb{R}^{N\times N}$ an orthonormal two-dimensional block discrete cosine transform. The group-of-pictures transform is $\Psi = \Psi_{\text{t}} \otimes \Psi_{\text{s}} \in \mathbb{R}^{k\times k}$, and the coefficient vector of group g is $\boldsymbol{c}_g = \Psi \boldsymbol{x}_g$ with components $c_{g,i}$, $i = 1, \dots, k$.

Lemma 2 (Distortion invariance)**.** *For $\boldsymbol{x}, \hat{\boldsymbol{x}} \in \mathbb{R}^k$ with $\boldsymbol{c} = \Psi\boldsymbol{x}$, $\hat{\boldsymbol{c}} = \Psi\hat{\boldsymbol{x}}$, $d(\boldsymbol{x}, \hat{\boldsymbol{x}}) = k^{-1}\|\boldsymbol{c} - \hat{\boldsymbol{c}}\|^2$.*

Proof. $\Psi^\top\Psi = I$ gives $\|\boldsymbol{x} - \hat{\boldsymbol{x}}\|^2 = \|\Psi(\boldsymbol{x} - \hat{\boldsymbol{x}})\|^2$. □

No motion search, rate–distortion optimisation, or entropy-coder state enters the transform stage; it is a fixed linear map.

4.2 Statistical Model

The design and the equivalence scale of Section 3 use the following model. It is a modelling assumption, not a claim about natural video; its parameters are defined operationally below and are estimated from data in the evaluation sections.

A1 The transform is that of Definition 4.

A2 Given the variance field, the pairs (coefficient, side information) are jointly Gaussian and independent across the index i.

A3 Each coefficient follows a first-order Gauss–Markov law across groups of pictures, $c_{g,i} = a_i\, c_{g-1,i} + w_{g,i}$, $w_{g,i} \sim \mathcal{N}(0, \omega_i)$ independent of the past, $|a_i| < 1$, with stationary variance $\lambda_i = \omega_i/(1 - a_i^2)$. Here a_i is the loop gain of the motion-compensated prediction and ω_i the variance of the motion-compensated innovation under a perfect reference.

A4 The receiver forms the side information $z_{g,i} = a_i\, \hat{c}_{g-1,i}$ from the conditional-mean reconstruction $\hat{c}_{g-1,i}$ of the previous group.

A5 The parameters $(\lambda_i, \omega_i, a_i)$ are known to both ends at subband granularity, transmitted as protected metadata.

4.3 Receiver-Side Prediction as Scalar Kalman Filtering

Lemma 3 (Side-information quality recursion)**.** *Under A2–A4 let $D_i(g-1) = \mathbb{E}(c_{g-1,i} - \hat{c}_{g-1,i})^2$. Then $c_{g,i} - z_{g,i}$ is zero-mean Gaussian with variance*

$$\mu_i(g) = \omega_i + a_i^2\, D_i(g-1), \tag{9}$$

and $z_{g,i}$ is the linear minimum mean-squared-error predictor of $c_{g,i}$ from the past.

Proof. For a conditional-mean estimate of a Gaussian variable the error $e_{g-1,i} = c_{g-1,i} - \hat{c}_{g-1,i}$ is independent of $\hat{c}_{g-1,i}$. Moreover $e_{g-1,i}$ is a function of the past and of earlier noise, hence independent of $w_{g,i}$. Then $c_{g,i} - z_{g,i} = a_i e_{g-1,i} + w_{g,i}$ has variance $a_i^2 D_i(g-1) + \omega_i$. Optimality: $\mathbb{E}[c_{g,i} z_{g,i}] = a_i^2 \mathbb{E}[\hat{c}_{g-1,i}^2] = \mathbb{E}[z_{g,i}^2]$, so the regression coefficient is one. □

Lemma 3 makes the receiver a bank of scalar Kalman filters [23]. Let $\eta \geq 0$ be the total precision (inverse variance) that the transmission of the current group adds about $c_{g,i}$, and let the update be $D_i(g) = \big(1/\mu_i(g) + \eta\big)^{-1}$.

Proposition 3 (Steady-state distortion)**.** *For constant $\eta > 0$ the recursion $D \mapsto \big(1/(\omega + a^2 D) + \eta\big)^{-1}$ has a unique fixed point*

$$D_\infty(\eta) = \frac{-B + \sqrt{B^2 + 4\eta a^2 \omega}}{2\eta a^2}, \qquad B = 1 + \eta\omega - a^2, \tag{10}$$

(for $a = 0$, $D_\infty = \omega/(1+\eta\omega)$), and the recursion converges to it from every $D(0) \geq 0$ at geometric rate at most a^2.

Proof. Put $m = \omega + a^2 D$ and $F(D) = m/(1+\eta m)$. F is increasing, with $F'(D) = a^2/(1+\eta m)^2 \leq a^2 < 1$, so F is a contraction on $[0, \infty)$ and has a unique fixed point by Banach's theorem. The fixed-point equation $D(1 + \eta m) = m$ expands to $\eta a^2 D^2 + (1 + \eta\omega - a^2)D - \omega = 0$, whose only positive root is (10). □

4.4 Reverse Water-Filling with Side Information

Proposition 4 (Information function of the model)**.** *Let $\mu_i = \mathrm{Var}(c_i \mid z_i)$ and assume A2. Then*

$$\mathcal{R}_g(D \mid \boldsymbol{Z}) = \sum_{i=1}^{k} \tfrac{1}{2} \log_2^+(\mu_i/\theta)\,, \qquad \sum_{i=1}^{k} \min\{\theta, \mu_i\} = kD, \tag{11}$$

where $\theta \geq 0$ is the water level, unique for $0 < D \leq k^{-1} \sum_i \mu_i$. The optimal per-coefficient distortions are $D_i = \min\{\theta, \mu_i\}$.

Proof. Converse. For $U - \boldsymbol{X} - \boldsymbol{Z}$ the chain rule and $\boldsymbol{Z} - \boldsymbol{X} - U$ give $I(\boldsymbol{X}; U) - I(\boldsymbol{Z}; U) = I(\boldsymbol{X}; U \mid \boldsymbol{Z}) = h(\boldsymbol{X} \mid \boldsymbol{Z}) - h(\boldsymbol{X} \mid U, \boldsymbol{Z})$. By A2, $h(\boldsymbol{X} \mid \boldsymbol{Z}) = \sum_i \frac{1}{2} \log_2(2\pi e \mu_i)$. Subadditivity gives $h(\boldsymbol{X} \mid U, \boldsymbol{Z}) \leq \sum_i h(c_i \mid U, \boldsymbol{Z})$, and the maximum-entropy property gives $h(c_i \mid U, \boldsymbol{Z}) \leq \frac{1}{2} \log_2(2\pi e D_i)$, with $D_i = \mathbb{E}(c_i - f_i(U, \boldsymbol{Z}))^2$. Each difference $h(c_i \mid z_i) - h(c_i \mid U, \boldsymbol{Z})$ equals $I(c_i; U, \boldsymbol{Z}_{-i} \mid z_i) \geq 0$, so it is at least $\frac{1}{2} \log_2^+(\mu_i/D_i)$. Hence $\mathcal{R}_g(D \mid \boldsymbol{Z}) \geq \min\{\sum_i \frac{1}{2} \log_2^+(\mu_i/D_i) : \sum_i D_i \leq kD\}$. The right side is a convex programme; its Karush–Kuhn–Tucker conditions give $D_i = \min\{\theta, \mu_i\}$ and (11) [12, §10.3.3].

Achievability. For $\mu_i > \theta$ take $U_i = c_i + N_i$ with N_i independent Gaussian of variance $\theta\mu_i/(\mu_i - \theta)$, and $f_i = \mathbb{E}[c_i \mid U_i, z_i]$. Then $U_i - c_i - z_i$ holds, $I(c_i; U_i \mid z_i) = \frac{1}{2} \log_2(\mu_i/\theta)$, and the distortion is θ. For $\mu_i \leq \theta$ send nothing and use $f_i = z_i$; the distortion is μ_i. Independence across i makes the rates add. □

Remark 4. For jointly Gaussian pairs the Wyner–Ziv information function equals the conditional rate–distortion function [9], as the proof shows. For non-Gaussian innovations the Gaussian expression bounds the conditional function from above, and the rate loss of the Wyner–Ziv problem relative to the conditional function is below 0.5 bit per sample for squared error [24].

4.5 State-Independent Reference Scale

Chunk 1's credits distortion by the rate required *given the receiver's actual side information*; at equal distortion, better side information then gives *less* credit. We replace it by a scale that depends only on the source and on the distortion.

For $D \in (0, k^{-1} \sum_i \lambda_i]$ let $\theta(D)$ be the unique solution of $\sum_i \min\{\theta, \lambda_i\} = kD$.

Definition 5 (Reference and intra information functions)**.**

$$\mathcal{R}_g^{\mathrm{ref}}(D) = \sum_{i=1}^{k} \tfrac{1}{2} \log_2^+ \left(\frac{\omega_i + a_i^2\, \theta}{\theta} \right), \qquad \theta = \theta(D), \tag{12}$$

$$\mathcal{R}_g^{\mathrm{intra}}(D) = \sum_{i=1}^{k} \tfrac{1}{2} \log_2^+ \left(\frac{\lambda_i}{\theta} \right), \qquad \theta = \theta(D). \tag{13}$$

Definition 6 (Rate–distortion-equivalent throughput, reference form)**.** $\mathcal{T}_{\mathrm{eq}}(g,h) = \mathcal{R}_g^{\mathrm{ref}}\big(D_g(h)\big)/T_g$.

Corollary 2 (Operational meaning)**.** $\mathcal{R}_g^{\mathrm{ref}}(D) = \mathcal{R}_g(D \mid \boldsymbol{Z}^{\mathrm{ref}})$*, where* $\boldsymbol{Z}^{\mathrm{ref}}$ *has conditional variances* $\mu_i^{\mathrm{ref}} = \omega_i + a_i^2 \min\{\theta(D), \lambda_i\}$*. That is,* $\mathcal{R}_g^{\mathrm{ref}}$ *is the rate of an ideal zero-delay predictive coder whose reference frame has the target distortion;* $\mathcal{R}_g^{\mathrm{intra}}$ *is the rate of a coder that uses no previous group.*

Proof. For $\lambda_i > \theta$ we have $\mu_i^{\mathrm{ref}} = \omega_i + a_i^2\theta \geq \theta$, so (11) gives $D_i = \theta$. For $\lambda_i \leq \theta$, $\mu_i^{\mathrm{ref}} = \omega_i + a_i^2\lambda_i = \lambda_i \leq \theta$, so $D_i = \lambda_i$ and the term vanishes. The water level therefore solves $\sum_i \min\{\theta, \lambda_i\} = kD$, and (11) reduces to (12). □

Proposition 5 (Converse in surplus form)**.** *Let* $S_g = \big[\mathcal{R}_g^{\mathrm{ref}}(D_g(h)) - \mathcal{R}_g(D_g(h) \mid \boldsymbol{Z}_g)\big]^+$ *be the* side-information surplus*. Then*

$$\mathcal{T}_{\mathrm{eq}}(g,h) \leq \frac{C_n(h) + S_g}{T_g}, \tag{14}$$

and $S_g = 0$ *whenever* $\mu_i(g) \geq \mu_i^{\mathrm{ref}}$ *for all* i*.*

Proof. By definition of S_g, $\mathcal{R}_g^{\mathrm{ref}} \leq \mathcal{R}_g(\cdot \mid \boldsymbol{Z}_g) + S_g$, and Proposition 2 bounds the first term of the right side by $C_n(h)$. For the second claim, (11) is non-decreasing in each μ_i at fixed D: raising μ_i keeps every allocation $\{D_i \leq \mu_i\}$ feasible and raises each term $\frac{1}{2}\log_2^+(\mu_i/D_i)$. Hence $\mathcal{R}_g(D \mid \boldsymbol{Z}_g) \geq \mathcal{R}_g(D \mid \boldsymbol{Z}^{\mathrm{ref}}) = \mathcal{R}_g^{\mathrm{ref}}(D)$. □

Corollary 3 (Necessary condition for the floor, reference form)**.** *If* $\mathbb{P}[\mathcal{T}_{\mathrm{eq}}(g,h) \geq T_{\min}] \geq 1 - \epsilon$ *and* $S_g \leq S_{\max}$ *almost surely, then* $\mathbb{P}\big[(C_n(h) + S_{\max})/T_g \geq T_{\min}\big] \geq 1 - \epsilon$*. In steady state,* $S_{\max} = 0$*.*

4.6 The Price of a Side-Information-Free Floor

Proposition 6 (Price of firewalling)**.** *For* $\theta = \theta(D)$*,*

$$\pi_g(D) = \mathcal{R}_g^{\mathrm{intra}}(D) - \mathcal{R}_g^{\mathrm{ref}}(D) = \sum_{i:\lambda_i > \theta} \tfrac{1}{2} \log_2 \frac{\lambda_i}{\omega_i + a_i^2\theta} \;\geq 0, \tag{15}$$

with equality if and only if $a_i = 0$ *for every* i *with* $\lambda_i > \theta$*.*

Proof. For $\lambda_i > \theta$ and $a_i \neq 0$, $\omega_i + a_i^2\theta < \omega_i + a_i^2\lambda_i = \lambda_i$. Terms with $\lambda_i \leq \theta$ vanish in both functions. □

Table 5 evaluates the per-coefficient price for a coded coefficient with $\lambda/\theta = 100$.

5 Architecture: Isolated Digital Floor and Linear Analogue Refinement

5.1 Overview

Each group of pictures passes through the following stages; Table 6 lists the mathematical object attached to each.

Table 5: Rates in bits per coded coefficient per group of pictures at $\lambda/\theta = 100$ (so $\omega = (1-a^2)\lambda$): intra rate $\frac{1}{2}\log_2(\lambda/\theta)$, reference rate $\frac{1}{2}\log_2((\omega + a^2\theta)/\theta)$, firewall price (15), and the price as a fraction of the intra rate. Values were checked numerically.

a	ω/λ	Intra	Reference	Price	Price / Intra
0.50	0.7500	3.322	3.117	0.205	0.062
0.80	0.3600	3.322	2.598	0.724	0.218
0.90	0.1900	3.322	2.154	1.168	0.352
0.95	0.0975	3.322	1.707	1.615	0.486
0.99	0.0199	3.322	0.785	2.537	0.764

Table 6: Modules of the architecture.

Module	Operation	Mathematical object
Transform	$\boldsymbol{c}_g = \Psi \boldsymbol{x}_g$, fixed and orthonormal	Lemma 2
Variance field	Protected metadata $(\lambda_i, \omega_i, a_i)$ per subband	Gauss–Markov parameters, A3
Floor layer	Digital block of the coefficients with $\lambda_i > \theta_F$; no dependence on earlier groups	θ_F, $\mathcal{R}_g^{\text{intra}}$, Proposition 6
Refinement layer	Linear analogue transmission of the floor residual	Gains from Proposition 8
Receiver-side prediction	Motion-compensated prediction of the previous reconstruction; scalar Kalman recursion	Lemma 3, Proposition 3
Decoder	Conditional mean of every coefficient	Proposition 9
Resource split	n_F channel uses for the floor, $n_A = n - n_F$ for refinement	Chunks 3–4

Encoder (per group).

E1. Compute $\boldsymbol{c}_g = \Psi \boldsymbol{x}_g$.

E2. Estimate the variance field from the source frames only (open loop); quantise it as metadata.

E3. Solve $\mathcal{R}_g^{\text{ref}}(D(\theta_F)) = T_{\min} T_g$ for θ_F and set $\mathcal{F} = \{i : \lambda_i > \theta_F\}$.

E4. Quantise and entropy-code $\{c_{g,i}\}_{i \in \mathcal{F}}$ with a static model from the variance field; append metadata; protect with a code whose rate and outage level are fixed by chunk 3. No quantity from an earlier group is used.

E5. Form the residual $r_i = c_{g,i} - \hat{c}_i^{\text{F}}$ for $i \in \mathcal{F}$ and $r_i = c_{g,i}$ otherwise.

E6. Choose the gains of Proposition 8 from the model recursion (9) and the channel-quality estimate, and the active set $\mathcal{A}$.

E7. Spread across packets by an orthogonal matrix, map to n_A complex channel uses.

Decoder (per group).

D1. Estimate the channel and equalise; this fixes the per-dimension noise variances ν_i.

D2. Decode the floor block and record the success flag.

D3. Form $\boldsymbol{z}_g = \Psi\,\mathrm{MC}(\hat{\boldsymbol{x}}_{g-1})$ by decoder-side motion compensation; when the floor block decoded, its low-resolution reconstruction guides the motion search, in the manner of hash-based motion compensation at the receiver [25]. No motion vectors are transmitted.

D4. Compute the conditional mean of Proposition 9.

D5. Apply $\Psi^\top$ and update the recursion state $D_i(g)$.

5.2 Floor Layer

Definition 7 (Floor level)**.** θ_F is the unique solution in $(0, \max_i \lambda_i]$ of $\mathcal{R}_g^{\mathrm{ref}}(D(\theta)) = T_{\min} T_g$, and $D_F = D(\theta_F) = k^{-1} \sum_i \min\{\theta_F, \lambda_i\}$. The floor payload per group is

$$B_F = \mathcal{R}_g^{\mathrm{intra}}(D_F) + |\mathcal{F}|\,\Delta_q + B_{\mathrm{meta}}, \tag{16}$$

where $\Delta_q \geq 0$ is the operational overhead per coded coefficient of scalar quantisation and B_{meta} the metadata size.

For entropy-coded uniform scalar quantisation, $\Delta_q \to \frac{1}{2}\log_2(\pi e/6) \approx 0.2546$ bit in the high-resolution limit [26]; at the rates of Table 1 most coefficients are below the water level, and Δ_q is a measured parameter of the evaluation. By Proposition 6 and Definition 7,

$$B_F = T_{\min} T_g + \pi_g(D_F) + |\mathcal{F}|\,\Delta_q + B_{\mathrm{meta}}, \tag{17}$$

so the floor must carry the credited $T_{\min} T_g$ bits plus the firewall price, the quantiser overhead, and the metadata.

Proposition 7 (Floor isolation)**.** *Assume the ideal Gaussian test channel of Proposition 4 for the floor layer. If the floor block is decoded, then the conditional-mean decoder of Proposition 9 attains $D_i \leq \min\{\theta_F, \lambda_i\}$ for every coefficient, whatever the quality of the side information and of the analogue layer, and hence $\mathcal{T}_{\mathrm{eq}}(g, h) \geq T_{\min}$.*

Proof. The floor observation alone leaves error variance $\min\{\theta_F, \lambda_i\}$. The conditional mean given additional observations cannot have larger error variance than the conditional mean given a subset. Thus $D_g(h) \leq D_F$. $\mathcal{R}_g^{\mathrm{ref}}$ is non-increasing in D, so $\mathcal{T}_{\mathrm{eq}} \geq \mathcal{R}_g^{\mathrm{ref}}(D_F)/T_g = T_{\min}$. □

Remark 5. A floor that used the previous reconstruction would need only $\mathcal{R}_g^{\mathrm{ref}}$ bits, but a single outage would degrade the side information of the next group, which is the error-propagation mechanism of closed-loop prediction. Proposition 7 removes this coupling at the price $\pi_g(D_F)$ of Table 5; the price is paid by the floor layer only, while the refinement layer exploits side information. If the decoder detects that analogue or side-information quality is mismatched, it falls back to the floor reconstruction, which preserves the guarantee.

Corollary 4 (Necessary capacity of the floor)**.** *Let n_F channel uses carry the floor block. Any scheme delivering B_F bits with block-error probability at most ϵ_F requires $\mathbb{P}[C_{n_F}(h) \geq B_F] \geq 1 - \epsilon_F$. Chunk 3 sharpens this by the dispersion bound [11].*

5.3 Refinement Layer

After the floor, component i has transmitted variance $\vartheta_i = \min\{\lambda_i, \theta_F\}$ and, at the receiver, posterior variance

$$\kappa_i = \Big(\frac{1}{\mu_i} + \Big[\frac{1}{\theta_F} - \frac{1}{\lambda_i}\Big]^+\Big)^{-1} \leq \vartheta_i, \tag{18}$$

since the floor adds the precision $1/\theta_F - 1/\lambda_i$ to the prior given by the prediction. The refinement layer sends $s_i = g_i r_i$ on one real dimension and the receiver sees noise of variance ν_i after equalisation.

Proposition 8 (Optimal linear allocation with side information)**.** *Minimise* $\sum_i D_i$ *with* $D_i = (1/\kappa_i + x_i/\nu_i)^{-1}$, $x_i = g_i^2 \geq 0$, *subject to* $\sum_i \vartheta_i x_i \leq \mathcal{P}$. *The unique optimum has, for a multiplier* $\xi > 0$,

$$D_i = \min\left\{\kappa_i,\ \sqrt{\xi\,\nu_i\,\vartheta_i}\right\}, \qquad P_i = \vartheta_i x_i = \left[\sqrt{\nu_i\vartheta_i/\xi} - \nu_i\vartheta_i/\kappa_i\right]^+, \tag{19}$$

active ($x_i > 0$) *exactly when* $\kappa_i^2 > \xi\nu_i\vartheta_i$, *and* ξ *is the unique root of* $\sum_i P_i(\xi) = \mathcal{P}$.

Proof. $x \mapsto (1/\kappa + x/\nu)^{-1}$ is convex and the constraint is linear, so the Karush–Kuhn–Tucker conditions are necessary and sufficient. Stationarity reads $-D_i^2/\nu_i + \xi\vartheta_i - \varsigma_i = 0$ with $\varsigma_i \geq 0$ and $\varsigma_i x_i = 0$. If $x_i > 0$, $D_i = \sqrt{\xi\nu_i\vartheta_i} < \kappa_i$; if $x_i = 0$, $D_i = \kappa_i$ and $\kappa_i \leq \sqrt{\xi\nu_i\vartheta_i}$. Solving $x_i = \nu_i(1/D_i - 1/\kappa_i)$ gives P_i. $\sum_i P_i(\xi)$ is continuous, strictly decreasing where positive, tends to ∞ as $\xi \downarrow 0$ and is 0 for $\xi \geq \max_i \kappa_i^2/(\nu_i\vartheta_i)$. □

Corollary 5 (Special cases and redundancy ratio)**.** *For* $\kappa_i = \vartheta_i = \lambda_i$ *(no side information, no floor),* (19) *is the linear-transform power allocation of SoftCast-type schemes [16], with gains* $g_i^2 \propto \lambda_i^{-1/2}$ *at high signal-to-noise ratio. The ratio* $\varrho_i = \vartheta_i/\kappa_i \geq 1$ *is the factor by which a linear layer over-spends power on the part of* r_i *the receiver already predicts; an encoder with the same prediction would spend* κ_i *in place of* ϑ_i.

Remark 6. If the number of active components is limited to $|\mathcal{A}| \leq 2n_A$, the problem is a mixed-integer convex programme. A Lagrangian price per active component selects those whose saving $\kappa_i - \min_x[D_i(x) + \xi\vartheta_i x]$ exceeds the price; the duality gap is at most the largest single saving. The gains use the model recursion (9) with the *expected* channel quality (open loop), so no receiver feedback or encoder-side prediction loop enters; robustness to mismatch is deferred to chunk 4.

5.4 Decoder

Proposition 9 (Conditional-mean decoder)**.** *Under A2–A4 and the ideal floor test channel, with* $\hat{c}_i^{\mathrm{F}} = 0$ *for* $i \notin \mathcal{F}$, *the conditional mean of* $c_{g,i}$ *given the prediction* z_i, *the floor reconstruction* $\hat{c}_i^{\mathrm{F}}$ *and the equalised observation* $y_i = g_i r_i + n_i$, $n_i \sim \mathcal{N}(0, \nu_i)$, *is*

$$\hat{c}_{g,i} = D_i\Big[\frac{z_i}{\mu_i} + \Big(\frac{1}{\theta_F} + \frac{g_i^2}{\nu_i}\Big)\hat{c}_i^{\mathrm{F}} + \frac{g_i}{\nu_i}\,y_i\Big], \qquad \frac{1}{D_i} = \frac{1}{\kappa_i} + \frac{g_i^2}{\nu_i}, \tag{20}$$

with error variance D_i *(for* $i \notin \mathcal{F}$ *the* $1/\theta_F$ *term is absent).*

Proof. Model the floor as $U = c + N_f$ with N_f independent of everything and variance $\theta_F\lambda_i/(\lambda_i - \theta_F)$; then $\hat{c}^{\mathrm{F}} = (1 - \theta_F/\lambda_i)U$ and the observation U has precision $1/\theta_F - 1/\lambda_i$. Combining the prior $\mathcal{N}(z_i, \mu_i)$ with U gives mean $\kappa_i(z_i/\mu_i + \hat{c}_i^{\mathrm{F}}/\theta_F)$ and variance κ_i. The observation $y_i/g_i + \hat{c}_i^{\mathrm{F}}$ of $c_{g,i}$ has noise variance ν_i/g_i^2 and is independent of the previous information; precision-weighted combination gives (20). □

Proposition 9 with $\eta_i = 1/\theta_F - 1/\lambda_i + g_i^2/\nu_i$ is the measurement update of Proposition 3; the steady-state distortion of the composite system is therefore $D_\infty(\eta_i)$ per coefficient.

5.5 Upfront Compression of the Composite Encoder

Proposition 10 (Pre-channel distortion fraction)**.** *Let* $\mathcal{U} = \{i : g_i = 0\}$. *Then* $D_{\mathrm{enc}} = k^{-1}\sum_{i\in\mathcal{U}}\kappa_i$, *and with the allocation of Proposition 8,*

$$\beta_{\mathrm{up}} = \frac{\sum_{i\in\mathcal{U}} D_i}{\sum_{i=1}^{k} D_i} \in [0, 1], \tag{21}$$

that is, the share of end-to-end distortion contributed by the truncated *components;* $\beta_{\mathrm{up}} = 0$ *if and only if* $\mathcal{U} = \varnothing$.

Proof. For $i \notin \mathcal{U}$ the noiseless analogue symbol returns r_i exactly, which together with $\hat{c}_i^{\mathrm{F}}$ gives $c_{g,i}$ exactly. For $i \in \mathcal{U}$ the best noiseless reconstruction uses only z_i and the floor, with error variance κ_i. Proposition 8 gives $D_i = \kappa_i$ on $\mathcal{U}$. □

The floor quantiser is not counted in D_{enc} because its error travels in the refinement layer; this contrasts with a fixed-rate digital chain (Proposition 1).

5.6 Positioning

DCAST sends modulo-reduced (coset) coefficients through soft transmission, sets the coset step from the noise of an encoder-side prediction, and performs decoder-side motion estimation [27,28]; coset codes originate with syndrome-based distributed source coding [29]. A modulo operation introduces a wrap-around failure whose probability depends on the step relative to the prediction-plus-channel error; the present refinement layer avoids that non-linearity at the price of the redundancy ratio ϱ_i, and a chance-constrained modulo extension is deferred. Systematic lossy coding [30] sends the source uncoded and a Wyner–Ziv description that uses the noisy analogue output as side information, which reverses our ordering: the digital description then depends on analogue quality and carries no probabilistic guarantee. Hybrid digital–analogue broadcast codes with side information [31] are Gaussian-source results without temporal recursion or a throughput floor. In the surveyed works we find no counterpart of Proposition 7 or Proposition 6.

6 The Floor Block at Finite Blocklength

This section replaces Corollary 4 by a certified bound on the block-error probability of the floor layer. Throughout, $Q(x) = \int_x^\infty (2\pi)^{-1/2} e^{-t^2/2}\,\mathrm{d}t$ and information quantities are in bits. Two findings organise the section: at a floor of 2 Mbps the finite-blocklength penalty is a small fraction of the channel uses (Table 7), whereas fading outage is not, and the floor guarantee is therefore governed by diversity (Table 9).

6.1 Channel, Ensemble, and Information Density

The floor block of group g occupies $n_F = Lm$ complex channel uses arranged in L blocks of m uses, with gain h_ℓ constant within block ℓ, known to the receiver and unknown to the transmitter. The block-level signal-to-noise ratio is $\gamma_\ell = |h_\ell|^2 P_F/\nu$. The codeword is interleaved across all L blocks. The codebook ensemble has M codewords with entries independent and identically distributed (i.i.d.) $\mathcal{CN}(0, P_F)$. The information density, with the receiver output taken to be $(\boldsymbol{Y}, h)$, is

$$\imath_{\mathrm{F}}(\boldsymbol{x}; \boldsymbol{y} \mid h) = \sum_{\ell=1}^{L} \sum_{j \in \ell} \Big[\log_2(1+\gamma_\ell) + \log_2 e \Big(\frac{|y_j|^2}{\nu(1+\gamma_\ell)} - \frac{|y_j - h_\ell x_j|^2}{\nu} \Big) \Big]. \tag{22}$$

Lemma 4 (Moments of the information density)**.** *Conditionally on h,*

$$\mathbb{E}[\imath_{\mathrm{F}} \mid h] = m \sum_{\ell} \log_2(1+\gamma_\ell) =: I_F(h), \tag{23}$$

$$\mathrm{Var}[\imath_{\mathrm{F}} \mid h] = m \sum_{\ell} V_{\mathrm{iid}}(\gamma_\ell), \qquad V_{\mathrm{iid}}(\gamma) = \frac{2\gamma}{1+\gamma} \log_2^2 e = V(\gamma) + \Big(\frac{\gamma}{1+\gamma}\Big)^2 \log_2^2 e, \tag{24}$$

where $V(\gamma) = \frac{\gamma(\gamma+2)}{(1+\gamma)^2} \log_2^2 e$ is the variance conditional on a codeword of energy exactly P_F per use.

Proof. Normalise $\nu = 1$, $P_F = 1$ and drop the block index. With $s = \sqrt{\gamma}\, x$ and $y = s + z$, the bracket in (22) (in nats) is $\ln(1+\gamma) + |y|^2/(1+\gamma) - |z|^2$, and $|y|^2/(1+\gamma) - |z|^2 = \text{const} + 2\,\mathrm{Re}(s^* z)/(1+\gamma) - \gamma|z|^2/(1+\gamma)$. Given x, the variance is $(2|s|^2+\gamma^2)/(1+\gamma)^2$ because $\mathrm{Var}\,|z|^2 = 1$ and the cross term vanishes by symmetry; at $|x|^2 = 1$ this is $\gamma(\gamma+2)/(1+\gamma)^2$, and its average over x is the same. The conditional mean is $\ln(1+\gamma) - 1 + (\gamma|x|^2+1)/(1+\gamma)$, whose variance over x is $\gamma^2/(1+\gamma)^2$. The law of total variance gives $2\gamma/(1+\gamma)$. Conversion from nats^2 to bits^2 multiplies by $\log_2^2 e$. The sum over uses is a sum of independent terms given h. □

6.2 Certified Floor Guarantee

Let $\imath_{\mathrm{F}}$ be as in (22) and, for $M = 2^B$,

$$\epsilon_{\mathrm{DT}}(B; n_F, P_F) = \mathbb{E}\Big[2^{-\left[\imath_{\mathrm{F}} - \log_2 \frac{M-1}{2}\right]^+}\Big], \tag{25}$$

the expectation being over the fading h, the i.i.d. codeword and the noise.

Theorem 1 (Certified floor block). If $\epsilon_{\mathrm{DT}}(B_F + 1; n_F, P_F) \le \epsilon_F/2$, there is a code with 2^{B_F} messages whose *maximal* block-error probability, averaged over fading and noise, is at most ϵ_F. The bound holds for any distribution of the payload.

Proof. The dependence-testing bound [11, Theorem 17], applied to the channel $\boldsymbol{X} \mapsto (\boldsymbol{Y}, h)$ with input law the i.i.d. ensemble, gives a code with $M = 2^{B_F+1}$ messages and average error probability at most $\epsilon_{\mathrm{DT}}(B_F + 1; n_F, P_F) \le \epsilon_F/2$. Discarding the half of the codewords with the largest error probability leaves 2^{B_F} codewords, each with error probability at most twice the average, that is, at most ϵ_F. □

Remark 7 (Power constraint and integrity). The ensemble meets the power constraint in expectation. The energy of a drawn codeword fluctuates about $n_F P_F$ with relative standard deviation $n_F^{-1/2}$; discarding codewords above $(1+3n_F^{-1/2})n_F P_F$ removes about 0.14% of them and a back-off of $3n_F^{-1/2}$ in P_F restores the budget, which is below 0.04 dB for $n_F > 10^5$. A cyclic redundancy check of r bits inside the payload turns undetected decoding errors into detected ones except with probability 2^{-r}, which is absorbed into ϵ_F.

Lemma 5 (Outage plus margin). *For every $t \ge 0$, $\epsilon_{\mathrm{DT}}(B; n_F, P_F) \le \mathbb{P}\big[\imath_{\mathrm{F}} \le \log_2 \frac{M-1}{2} + t\big] + 2^{-t}$.*

Proof. On $\{\imath_{\mathrm{F}} \le \log_2 \frac{M-1}{2} + t\}$ the integrand is at most one; on the complement it is at most 2^{-t}. □

With $t = \log_2(4/\epsilon_F)$ the second term equals $\epsilon_F/4$, about 12 bits of margin at $\epsilon_F = 10^{-3}$, negligible against $B_F \sim 10^5$: the guarantee is an outage statement about $\imath_{\mathrm{F}}$.

Proposition 11 (Sampling certificate). *Let $\epsilon_1, \dots, \epsilon_N$ be independent draws of the integrand of* (25) *(values in $[0,1]$) with sample mean $\hat{\epsilon}$, and let $\mathrm{KL}(p\|q) = p \ln \frac{p}{q} + (1-p)\ln\frac{1-p}{1-q}$. Define $\bar{\epsilon}_\delta = \max\{q \ge \hat{\epsilon} :\ N\,\mathrm{KL}(\hat{\epsilon}\|q) \le \ln(1/\delta)\}$. Then $\mathbb{P}\big[\epsilon_{\mathrm{DT}} \le \bar{\epsilon}_\delta\big] \ge 1 - \delta$.*

Proof. For variables in $[0,1]$ with mean p, the lower-tail bound $\mathbb{P}[\hat{\epsilon} \le p - \tau] \le \exp(-N\,\mathrm{KL}(p-\tau\|p))$ holds [32]. If $p > \bar{\epsilon}_\delta$ then $\hat{\epsilon} < p$ and $\mathrm{KL}(\hat{\epsilon}\|p) > \ln(1/\delta)/N$, which the bound shows has probability at most δ. □

A floor is therefore *certified* when $\bar{\epsilon}_\delta \le \epsilon_F/2$ for $B = B_F + 1$; the sizing below supplies the starting point of the search.

Table 7: Relative excess channel uses ρ (percent) over B_F/C from (26), additive white Gaussian noise channel, maximal-error target ϵ_F. Each cell is "i.i.d. ensemble | equal-energy codewords". $B_F = 66{,}667$ bits is 2 Mbps over one frame ($G = 1$, 30 frames per second); $B_F = 533{,}333$ bits is $G = 8$.

B_F	ϵ_F	5 dB	10 dB	20 dB
66,667	10^{-3}	1.59 | 1.25	1.34 | 0.99	1.01 | 0.72
66,667	10^{-5}	2.14 | 1.68	1.81 | 1.33	1.36 | 0.96
533,333	10^{-3}	0.56 | 0.44	0.47 | 0.35	0.36 | 0.25
533,333	10^{-5}	0.75 | 0.59	0.63 | 0.47	0.48 | 0.34

Table 8: Validation of the sizing rule against the dependence-testing bound, $\gamma = 10$ dB. $\hat{\epsilon}$ estimates $\epsilon_{\mathrm{DT}}(B_F + 1)$ from 10^6 samples (6×10^5 in the second row); $\bar{\epsilon}$ is the upper confidence bound of Proposition 11 at $\delta = 10^{-6}$; the design target is $\epsilon_F/2$.

n_F	B_F	Sizing with	$\hat{\epsilon}$	$\bar{\epsilon}$	Target
1,000	3,300	V_{iid}	5.22×10^{-3}	5.61×10^{-3}	5×10^{-3}
1,000	3,342	V	2.98×10^{-2}	—	5×10^{-3}
154,897	533,333	V_{iid}	5.10×10^{-4}	6.38×10^{-4}	5×10^{-4}
154,706	533,333	V	7.48×10^{-3}	7.94×10^{-3}	5×10^{-4}

6.3 Sizing on the Additive White Gaussian Noise Channel

Let $C = \log_2(1+\gamma)$ and let V be $V_{\mathrm{iid}}(\gamma)$ for the i.i.d. ensemble certified by Theorem 1, or $V(\gamma)$ for equal-energy codewords (see Remark 8).

Proposition 12 (Normal-approximation sizing)**.** *The solution of* $n_F C - \sqrt{n_F V}\, Q^{-1}(\epsilon_F/2) = B_F$ *is*

$$n_F^{\mathrm{NA}} = \left(\frac{\sqrt{V}\, Q^{-1}(\epsilon_F/2) + \sqrt{V\, Q^{-1}(\epsilon_F/2)^2 + 4CB_F}}{2C} \right)^2, \tag{26}$$

and, with $q = Q^{-1}(\epsilon_F/2)$,

$$n_F^{\mathrm{NA}} = \frac{B_F}{C} + \frac{q\sqrt{VB_F}}{C^{3/2}} + \frac{Vq^2}{2C^2} + O(B_F^{-1/2}), \qquad \rho = \frac{n_F^{\mathrm{NA}} C}{B_F} - 1 \approx q\sqrt{\frac{V}{C\,B_F}} + \frac{Vq^2}{2CB_F}. \tag{27}$$

Proof. Put $x = \sqrt{n_F}$; the equation is $Cx^2 - \sqrt{V} q\, x - B_F = 0$ with the stated positive root. Expanding $\sqrt{a^2 + 4CB_F}$, $a = \sqrt{V}q$, to second order in $a^2/(4CB_F)$ and squaring gives (27). □

Because $B_F = T_{\min} T_g \propto G$, the relative overhead ρ decreases as $G^{-1/2}$: a longer group of pictures lowers the dispersion penalty at the price of latency. Table 7 evaluates ρ.

Remark 8 (Ensemble). Lemma 4 shows $V_{\mathrm{iid}} = V + (\gamma/(1+\gamma))^2 \log_2^2 e$: the excess is the variance of the codeword energy. The normal approximation with $V(\gamma)$ requires equal-energy codewords [11]; the dependence-testing bound with i.i.d. inputs does not certify it. Table 8 shows the consequence.

The V_{iid} rule meets its target within a few percent (the first row is 4% above it; certification adds a few channel uses), whereas the V rule misses by factors of 6 to 15 for i.i.d. inputs.

6.4 Block Fading and the Need for Diversity

For L independent Rayleigh blocks (power gains exponentially distributed with unit mean) and mean signal-to-noise ratio $\bar{\gamma}$, let $R_\epsilon(L, \bar{\gamma})$ be the ϵ-quantile of $L^{-1}\sum_\ell \log_2(1 + \bar{\gamma} X_\ell)$, the ϵ-outage rate per channel use. An ideal code needs $n_F^{\mathrm{out}} = B_F / R_{\epsilon_F}$ channel uses. For $L = 1$ the rate R_ϵ is a sharp proxy for the finite-blocklength optimum because the dispersion is zero [33]; for $L > 1$ we rely on the certificate of Proposition 11 rather than a converse.

Table 9: Fraction n_F/n of the channel uses of a group occupied by the floor: $W = 5$ MHz, $T_g = 8/30$ s ($n \approx 1.333 \times 10^6$), $B_F = 533{,}333$, $\bar{\gamma} = 10$ dB, $\epsilon_F = 10^{-3}$, Rayleigh blocks. Ideal outage-limited code B_F/R_{ϵ_F} versus the certified dependence-testing design (maximal error, $\bar{\epsilon} \le \epsilon_F/2$, $\delta = 10^{-6}$).

L	Ideal outage-limited	Certified
1	27.74	infeasible
2	1.409	infeasible
4	0.435	0.521
8	0.271	0.298
16	0.212	0.223
64	0.167	0.171
256	0.151	—
∞ (ergodic)	0.138	—

Table 10: Floor share n_F/n (percent) on the additive white Gaussian noise channel: $B_F = 533{,}333$, $T_g = 8/30$ s, $\epsilon_F = 10^{-3}$, i.i.d. ensemble. Entries near 100 leave no resource for refinement.

W [MHz]	5 dB	10 dB	20 dB
1	97.8	58.1	30.1
5	19.6	11.6	6.0
20	4.9	2.9	1.5

Corollary 6 (Diversity is a prerequisite)**.** *At the parameters of Table 9, an ideal code with $L \le 2$ needs more channel uses than the group has, and the floor is infeasible; the certified design needs $L \ge 4$ and occupies* 52%*,* 30%*,* 22% *and* 17% *of the channel uses for $L = 4, 8, 16, 64$.*

The diversity must come from frequency selectivity, antennas, or time variation within the group duration T_g; a static, flat link provides none of these.

6.5 Resource Consequences

Table 10 gives the floor's share of the channel uses on the additive white Gaussian noise channel. The remainder $n_A = n - n_F$ is available to the refinement layer, with analogue bandwidth ratio $b_A = 2n_A/k$ (two real dimensions per complex use). At $W = 5$ MHz and 10 dB the floor takes 11.6% and $b_A = 0.213$.

The floor-first split carries the energy constraint $n_F P_F + \mathcal{E}_A \le n\bar{P}$ with $\mathcal{E}_A$ the refinement energy of Proposition 8; boosting P_F shortens the floor but removes energy from refinement, and the joint optimum is the subject of the next chunk.

6.6 Segmentation and Rateless Alternatives

Proposition 13 (Segmentation penalty)**.** *Splitting the payload into K independently coded segments of B_F/K bits with maximal error ϵ_F/K each meets ϵ_F by the union bound, and the relative excess over B_F/C is $\rho_K \approx Q^{-1}(\epsilon_F/(2K))\sqrt{KV/(CB_F)}$.*

For $\gamma = 10$ dB, $B_F = 533{,}333$ and $\epsilon_F = 10^{-3}$ this gives 0.47% monolithically and 4.95% for $K = 64$, an increase of one order of magnitude; in a numerical search over the per-segment error level and an outer erasure-code redundancy, no redundancy improved on the union bound at this ϵ_F. Practical low-density parity-check codes have code blocks of the order of 10^4 bits, so the penalty applies unless the floor uses a long code or a rateless code. Under fading the union bound is conservative because the outage event is common to all segments.

Rateless floor. With stop feedback, a rateless code [20,21,34] fails exactly when the information accumulated by the deadline $n_F^{\max}$ is below the threshold, the same outage event as for a fixed-rate code, so the guarantee is unchanged; the gain is that the floor finishes early when the channel is good and the unused uses go to refinement. For an ideal rateless code (no dispersion, no margin, no feedback delay) with $n_F^{\max} = B_F/R_{\epsilon_F}$, the mean floor share falls from 43.5% to 14.6% for $L = 4$, from 21.3% to 13.9% for $L = 16$, and from 16.7% to 13.8% for $L = 64$, at the parameters of Table 9.

6.7 End-to-End Floor Guarantee

A6 Whenever the floor block is decoded, the floor source code attains $D_i \le \min\{\theta_F, \lambda_i\}$ for every coefficient; if it does so only with probability $1 - \epsilon_q$, replace ϵ_F below by $\epsilon_F + \epsilon_q$.

Theorem 2 (Chance-constrained floor). Under A1–A6, let (n_F, P_F) satisfy the condition of Theorem 1 for B_F of Definition 7. Then for every group g,

$$\mathbb{P}[\mathcal{T}_{\text{eq}}(g, h) < T_{\min}] \le \epsilon_F, \tag{28}$$

irrespective of the side information, of the refinement layer, and of the payload distribution.

Proof. By Theorem 1 the floor block fails with probability at most ϵ_F. If it decodes, Proposition 7 together with A6 gives $\mathcal{T}_{\text{eq}} \ge T_{\min}$. □

Corollary 7 (Violation rate)**.** *In stationary operation the long-run fraction of groups with $\mathcal{T}_{\text{eq}} < T_{\min}$ is at most ϵ_F. If channel realisations are independent across groups, violations are a Bernoulli process with mean interval T_g/ϵ_F: for $T_g = 8/30$ s this is* 26.7 *s,* 267 *s (*4.4 *min),* 2,667 *s (*44 *min) and* 26,667 *s (*7.4 *h) for $\epsilon_F = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$. Slow fading clusters violations without changing their long-run fraction.*

6.8 Open Items

The joint choice of (n_F, P_F) against the refinement energy and the peak-power constraint of Problem 2, and robustness of the open-loop gain design to model mismatch, are treated next.

7 Discussion

7.1 Measured Against the Paper's Own Metrics

The pilot of Section D can be re-expressed in the quantities defined in Sections 3 and 4. Table 11 gives the reference throughput $\mathcal{T}_{\text{eq}}$ (computed from the measured distortion of each group by Definition 6), the efficiency $\eta = \mathcal{T}_{\text{eq}}/C$ relative to the capacity $C = n_{\text{eff}} C(\gamma)/T_g$, and the pre-channel distortion fraction β_{up} (model distortions; Proposition 10). Rates in "equivalent" units are scaled to 720p25 at equal bits per sample (1 kbit/s in the pilot is 4.09×10^{-3} Mbps equivalent).

Three observations follow. First, the floor requirement holds in the reference metric: the floor-only reconstruction has $\mathcal{T}_{\text{eq}} = 489$ kbit/s (2.0 Mbps equivalent) in each of the 16 groups, and the measured minimum over groups is 2.17 Mbps equivalent or more at every signal-to-noise ratio. Second, the efficiency is nearly flat, $\eta \approx 0.65$ to 0.66: the design tracks the capacity but at two thirds of it, so the ten-megabit ceiling is reached only if the capacity is about $10/0.66 \approx 15$ Mbps equivalent; at the point where the capacity is 10 Mbps equivalent the design delivers 6.6. Third, the efficiency of the linear design is 0.02 to 0.12, which quantifies the gain of the hybrid structure at this bandwidth ratio in the paper's own scale.

The pre-channel distortion fraction is high: 0.78 to 0.84 on average for the proposed design, against 0.06 to 0.10 for the linear one. The reading is the following. At a bandwidth ratio

Table 11: Pilot results in the paper's own metrics, mean over 16 groups (minimum or maximum over groups in parentheses). Capacity and $\mathcal{T}_{\text{eq}}$ in Mbps equivalent.

		Proposed (floor + side information)			Linear, no floor, no side information		
γ [dB]	Capacity	$\mathcal{T}_{\text{eq}}$ (min)	η	β_{up} (max)	$\mathcal{T}_{\text{eq}}$	η	β_{up}
6	6.70	4.34 (2.17)	0.648	0.776 (0.983)	0.151	0.023	0.077
8	8.30	5.46 (2.66)	0.659	0.777 (0.937)	0.344	0.041	0.064
10	10.00	6.65 (3.45)	0.665	0.802 (0.918)	0.727	0.073	0.074
12	11.78	7.76 (4.34)	0.659	0.837 (0.915)	1.361	0.116	0.100

Table 12: Claims and the evidence available for them.

Claim	Evidence	Status
Quality floor conditional on a decoded floor block	Proposition 7, Theorem 2; pilot: floor-only ≥ 31.6 dB and $\mathcal{T}_{\text{eq}} = 489$ kbit/s in all groups	Supported
Outage level of the floor block	Theorem 1 with certificate; Table 8; not exercised under fading in the pilot	Supported analytically
Analytic model predicts quality	Model-minus-measured 0.18 to 0.36 dB with floor and side information (Section D)	Supported on one clip
Gain over linear analogue at low bandwidth ratio	+8.1 to +10.2 dB; η of 0.66 against 0.02–0.12	Supported on one clip
Peak-signal-to-noise-ratio superiority over a digital codec	-3.3 to -3.5 dB against zero-latency H.264, about -7 dB against the default preset, both with an ideal code and oracle rate adaptation	**Not supported**
Minimal channel-blind compression	$\beta_{\text{up}} = 0.78$–0.84 at $b \approx 0.17$ (Table 11); reduction is forced by the converse	**Not supported**
Robustness to signal-to-noise-ratio uncertainty	Not simulated; the 3 dB margin experiment costs the proposed design 0.37 dB per dB against about 0.55 for zero-latency H.264, an extrapolated advantage that would close the gap only at margins of 15–20 dB, and the floor is infeasible at a 6 dB margin for the 10 dB nominal point	Untested
Encoder-complexity advantage	Not measured	Untested

of about 0.17 most coefficients cannot be transmitted at all (Proposition 2, Table 1), and the distortion of the truncated components dominates the end-to-end distortion of the proposed design because its channel noise is small; in the linear design the channel noise dominates, so the truncated share is small. A decoded digital chain has $\beta_{\text{up}} = 1$ by definition, since it has no channel noise after decoding. The metric therefore does *not* show that the proposed design compresses less in the sense of Proposition 1 at this bandwidth ratio; the reduction is forced by the converse and not by the architecture.

7.2 What the Evidence Supports

7.3 Where a Digital Codec Should Still Win

On the pilot the digital baselines win in peak signal-to-noise ratio, and the measured structural costs of the design explain most of the gap:

- the firewall price (Proposition 6): up to 35% of the intra rate at a prediction gain of 0.9 and 76% at 0.99 (Table 5);
- metadata: 36,360 bits per group, 12% of the floor block;

- a scalar quantiser with a static, zeroth-order code length, against context-adaptive entropy coding and motion compensation in the codec; the floor block is 1.90 times the credited payload and its quality per bit is about 3 dB below zero-latency H.264;
- the inefficiency of linear analogue transmission at a bandwidth ratio below one (Section 4): the analogue-only variant reaches 31.6 dB where the hybrid reaches 40.6 dB at 10 dB;
- the margin that the floor requires (Table 19).

We therefore expect a digital codec followed by a capacity-achieving code to be preferable when most of the following hold: the mean signal-to-noise ratio is known to within a few decibel at encoding time or the cost of a margin is acceptable; the delay budget allows long groups of pictures and bi-directional prediction; the bandwidth ratio is well below one; the source has low entropy or has been compressed before; and the encoder has no binding complexity or energy limit. The pilot satisfies most of these conditions, which is consistent with its outcome.

The conditions under which the hybrid design may be preferable are hypotheses, not findings: *(i)* no or very delayed feedback with large signal-to-noise-ratio uncertainty, where a digital design must carry a margin that exceeds that of the floor alone; *(ii)* fading with a hard per-group deadline, no retransmission, and a requirement stated as a chance constraint on delivered quality; *(iii)* bandwidth ratios near one, where the reduction required of the encoder is small and the linear structure is most efficient; *(iv)* encoders without motion search or entropy coding of the bulk of the data. None of them was tested.

7.4 Relation to the Original Requirement

The requirement was a throughput of at least 2 Mbps, rising to 10 Mbps, with little upfront compression. In the terms of this paper the first part is met conditionally: a 2 Mbps-equivalent quality is delivered with probability at least $1 - \epsilon$ provided the floor block is decoded and the channel provides the diversity of Corollary 6. The ceiling is met only in proportion to the capacity, at an efficiency of about 0.66. The second part is not met at the bandwidth ratio of the pilot: the converse (Proposition 2) forces a reduction of the source, and what the architecture saves is not the reduction but the encoder machinery of motion search and entropy coding for the analogue layer, a saving not measured here.

8 Limitations and Threats to Validity

Model. Gaussian coefficients, a first-order Gauss–Markov evolution per chunk, conditional-mean reconstruction and an ideal Gaussian test channel for the floor (A1–A6) are modelling assumptions. The pilot supports them on one clip only. The reference scale $\mathcal{T}_{\mathrm{eq}}$ uses a Gaussian surrogate of the information function, which bounds the true one from above.

Certificates. The floor certificate assumes i.i.d. Gaussian codebooks, independent Rayleigh blocks and perfect channel knowledge; clipping and estimation error are modelled for the refinement layer but not yet included in the certificate. Assumption A8 (monotonicity in the signal-to-noise ratio) was checked at one design point. The controller result is local.

Evidence. One clip, previously compressed, animated and downscaled; sixteen groups; additive white Gaussian noise; the floor sized at the true signal-to-noise ratio; a single noise realisation; encoder statistics computed per group from the source; a zero-motion side-information predictor.

Table 13: Phases required to test the remaining claims.

Phase	Deliverable	Acceptance criterion
1	Raw-sequence evaluation: at least six uncompressed sequences at 720p or above covering four motion classes; per-sequence statistics (λ, ω, a)	Model-minus-measured gap at most 0.5 dB in at least 80% of groups
2	Fading simulator with correlated blocks, delayed and noisy feedback; digital baselines with the same feedback and a realistic code	Report outage, availability and quality distributions; tests of hypotheses (i)–(iii)
3	Decoder-side motion-compensated side information	Side-information gain of at least 1 dB with the floor
4	Parametric variance field	Metadata at most 3% of the floor block
5	Context-adaptive floor coding	Floor block at most 1.3 times the credited payload
6	Certificate with clipping and estimation error; analogue fallback below floor feasibility	Certified error bound recomputed; fallback quality reported
7	Software-defined-radio test bench	Measured peak-to-average power ratio, signal-to-noise-and-distortion ratio, latency, and encoder operations per sample against H.264 on the same processor

Baselines.
Digital baselines are given an ideal capacity-achieving code and oracle rate adaptation, which favours them; the linear baseline is our implementation at chunk granularity; no learned baseline was run.

Metrics.
Peak signal-to-noise ratio only, no perceptual metric; $\mathcal{T}_{\text{eq}}$ and β_{up} computed from model distortions.

Implementation.
Encoder complexity, latency of a real implementation, and amplifier behaviour were not measured.

9 Conclusion

We formulated the transmission of video at a few megabits per second as a chance-constrained floor problem on a reference throughput scale, proved a converse and a floor-isolation result, derived closed-form allocations for the analogue layer under side information, estimation error and peak power, and certified the floor block at finite blocklength; at the target scale the dispersion penalty is below two percent of the channel uses and diversity governs feasibility. A pilot on one previously compressed clip supports the floor guarantee, the analytic model within 0.4 dB, and a gain of 8 to 10 dB over linear analogue transmission at a bandwidth ratio of 0.17. It does not support a quality advantage over a digital codec with an ideal code or a minimal channel-blind compression; the measured structural costs and the regime in which digital transmission should prevail are stated above. The case for the design rests on the guarantee, the analysis, and the untested hypotheses of Section 7, which the phases of Table 13 are designed to settle.

Table 14: Verification checks.

Quantity	Result
Allocation (19) versus constrained numerical optimiser (20 random instances)	maximum relative gap 8.1×10^{-11}
Riccati fixed point (10) versus iteration ($a = 0.9$, $\omega = 0.19$, $\eta = 3$)	0.164001746899620 both
Posterior variance (18) versus Gaussian conditioning ($\lambda = 1$, $\theta = 0.01$, $\mu = 0.2$)	0.009615384615 both
Information-density sampler ($m = 50$, $\gamma = 10$, 4×10^5 draws)	mean 119.90 (theory 119.89), variance 90.78 (theory 90.91)
Dependence-testing estimator versus sizing rule	Table 8
Ideal-rateless savings, certified fractions	Table 9 and Section 6

A Reference Implementation and Verification

The module `floorlab.py` implements the closed forms of the Kalman steady state (Proposition 3), the allocation solver (Proposition 8), the sizing rule (26), the dependence-testing estimator with the certificate of Proposition 11, the outage quantiles for L Rayleigh blocks, and an exact sampler of the information density. The sampler uses, for $a = (\sqrt{\gamma}x+z)/\sqrt{1+\gamma}$ and $b = (x-\sqrt{\gamma}z)/\sqrt{1+\gamma}$ (independent standard complex normal vectors of length m), the identities $z = (a-\sqrt{\gamma}b)/\sqrt{1+\gamma}$ and $\sum_j |y_j|^2/(1+\gamma) = \|a\|^2$; the sufficient statistics $(\|a\|^2, \|b\|^2, \mathrm{Re}\langle a, b\rangle)$ are drawn by Bartlett decomposition in $O(1)$ per block. Table 14 lists the checks.

B Joint Resource Allocation, Peak Power, and Robustness

B.1 Peak Power by Optimal Back-Off

?? 2 constrains the peak-to-average power ratio of the transmitted signal. Because the refinement symbols are continuous-valued and, after spreading and multicarrier modulation, approximately Gaussian, the ratio exceeds Λ with probability close to $1-(1-e^{-\Lambda})^{N_s}$ for N_s independent samples [35], so that $\Lambda \approx \ln(N_s/\delta)$: about 11.4 dB for $N_s = 1024$ and $\delta = 10^{-3}$. Enforcing that bound wastes power; we instead let the amplifier clip and account for the distortion.

Lemma 6 (Hard amplitude limiter)**.** *Let $x \sim \mathcal{CN}(0, \sigma^2)$ pass through $y = x$ for $|x| \le A$ and $y = Ax/|x|$ otherwise, and let $\Lambda = A^2/\sigma^2$. Then $y = \alpha x + d$ with d uncorrelated with x,*

$$\alpha = 1 - e^{-\Lambda} + \tfrac{\sqrt{\pi}}{2}\sqrt{\Lambda}\,\mathrm{erfc}(\sqrt{\Lambda}), \qquad \sigma_d^2 = \sigma^2(1 - e^{-\Lambda} - \alpha^2). \tag{29}$$

Proof. The envelope $r = |x|$ has density $(2r/\sigma^2)e^{-r^2/\sigma^2}$. Then $\mathbb{E}|y|^2 = \mathbb{E}\min\{r^2, A^2\} = \sigma^2(1 - e^{-\Lambda})$ and $\alpha\sigma^2 = \mathbb{E}[x^*y] = \mathbb{E}[r\min\{r, A\}] = \int_0^A r^2 f\,\mathrm{d}r + A\int_A^\infty r f\,\mathrm{d}r$. With $t = r^2/\sigma^2$ the first integral is $\sigma^2[1-(1+\Lambda)e^{-\Lambda}]$; integration by parts gives the second as $A\big[Ae^{-\Lambda} + \frac{\sqrt{\pi}}{2}\sigma\,\mathrm{erfc}(\sqrt{\Lambda})\big]$. Adding gives α. Defining $d = y - \alpha x$ makes $\mathbb{E}[x^*d] = 0$, and $\sigma_d^2 = \mathbb{E}|y|^2 - \alpha^2\sigma^2$. □

The decomposition is the classical Bussgang one [36]. We treat d as additive noise, an approximation: d is uncorrelated with x but not independent of it. After receiver equalisation by

Table 15: Optimal back-off Λ^* maximising (30), the resulting signal-to-noise-and-distortion ratio, and its shortfall against a distortion-free amplifier operated at the same back-off, for a peak signal-to-noise ratio ρ_{pk}.

ρ_{pk} [dB]	Λ^* [dB]	SNDR* [dB]	Shortfall [dB]
10	0.75	6.42	2.83
15	2.85	10.35	1.80
20	4.41	14.34	1.25
25	5.63	18.42	0.95
30	6.63	22.61	0.76

$1/\alpha$, the signal power $\sigma^2 = P_{\mathrm{pk}}/\Lambda$ meets noise $(\nu + \sigma_d^2)/\alpha^2$, so

$$\mathrm{SNDR}(\Lambda) = \frac{\alpha^2\,\rho_{\mathrm{pk}}}{(1 - e^{-\Lambda} - \alpha^2)\rho_{\mathrm{pk}} + \Lambda}, \qquad \rho_{\mathrm{pk}} = \frac{P_{\mathrm{pk}}}{\nu}, \tag{30}$$

with $\alpha = \alpha(\Lambda)$ from (29). The refinement layer of Proposition 8 uses the effective noise $\nu_i \leftarrow (\nu + \sigma_d^2)/\alpha^2$ (per real dimension, with the same scaling).

At $\rho_{\mathrm{pk}} = 20$ dB the optimal back-off is 4.4 dB rather than the 11.4 dB needed to avoid clipping, a gain of about 7 dB in transmitted average power. The closed forms (29) agree with Monte-Carlo simulation to 3×10^{-4} in α and 2×10^{-5} in σ_d^2/σ^2 (4×10^6 draws, $\Lambda = 1, 2, 4, 8$).

Revision of (P)**.** Replace the peak-to-average constraint by a hard limiter at amplitude $A_{\max}$ ($P_{\mathrm{pk}} = A_{\max}^2$) shared by both layers, with the clipping distortion of (29) entering the effective noise of each layer. For the floor, γ_ℓ is replaced by the corresponding signal-to-noise-and-distortion ratio; the certificate of Proposition 11 must then be recomputed with clipping included, which the sampler of Section A does not yet do.

B.2 Joint Choice of the Floor Power and the Channel-Use Split

The floor occupies n_F channel uses at per-use signal-to-noise ratio $\gamma_F = P_F/\nu_c$ (ν_c the complex noise power) and consumes the energy $E_F = n_F P_F$. Refinement receives $M = 2(n - n_F)$ real dimensions and the energy $\mathcal{P} = n\bar{P} - E_F$.

Problem 3 (Resource split)**.** Given B_F, ϵ_F and the refinement problem of Proposition 8 with a limit of M active components, choose γ_F to minimise the optimal refinement distortion $\Phi(\mathcal{P}, M)$, subject to $n_F(\gamma_F) \le n$ and the certificate of Theorem 1.

Lemma 7 (Saving of a component)**.** *At multiplier ξ a component with $\kappa > \sqrt{\xi\nu\vartheta}$ lowers the Lagrangian by*

$$s(\xi) = \kappa - \min_{x \ge 0} \left[D(x) + \xi\vartheta x \right] = \frac{\left(\kappa - \sqrt{\xi\nu\vartheta}\right)^2}{\kappa}. \tag{31}$$

Proof. At the minimiser $D = \sqrt{\xi\nu\vartheta}$ and $x = \nu(1/D - 1/\kappa)$, so the minimum is $D + \xi\vartheta\nu/D - \xi\vartheta\nu/\kappa = 2\sqrt{\xi\nu\vartheta} - \xi\nu\vartheta/\kappa$. □

At fixed ξ the best choice of at most M active components keeps those with the largest $s(\xi)$; the price ζ of one dimension is the saving of the marginal component (zero if fewer than M components are worth activating).

Proposition 14 (First-order optimality of the split)**.** *Let the floor use $n_F = B_F/C(\gamma_F)$ with $C = \log_2(1 + \gamma_F)$ (first order; dispersion neglected), hold the side information fixed, and let Φ*

Table 16: Floor power sweep on the synthetic model. $\mathcal{T}_{\text{eq}}$ is the reference throughput of Definition 6. The stationary point of (32) lies where the last two columns cross, between 9 and 9.5 dB.

γ_F [dB]	n_F/n	$\mathcal{T}_{\text{eq}}$ [Mbps]	$(1+\gamma_F)\ln(1+\gamma_F)-\gamma_F$	$2\zeta/(\xi\nu_c)$
6.0	0.333	9.204	4.02	21.53
8.0	0.269	9.616	8.23	14.58
9.0	0.244	9.723	11.65	12.29
10.0	0.223	9.776	16.38	10.40
10.5	0.214	9.781	19.37	9.55
11.0	0.205	9.773	22.87	8.75
12.0	0.189	9.714	31.74	7.25
14.0	0.164	9.375	60.10	4.49

denote the Lagrangian-dual optimal value, differentiable at the optimum, with $\partial\Phi/\partial\mathcal{P}=-\xi$ and $\partial\Phi/\partial M=-\zeta$. A stationary point in γ_F satisfies

$$(1+\gamma_F)\ln(1+\gamma_F)-\gamma_F=\frac{2\zeta}{\xi\,\nu_c}. \tag{32}$$

The left side increases strictly from 0, so the root is unique. If the dimension limit is slack ($\zeta=0$), the only root is $\gamma_F=0$, and the floor should occupy as many channel uses as the budget allows.

Proof. With $E_F=\nu_c B_F\gamma_F/C$ and $M=2(n-n_F)$, $\mathrm{d}\Phi/\mathrm{d}\gamma_F=\xi E_F'+2\zeta n_F'$. Using $n_F'=-B_FC'/C^2$, $E_F'=\nu_cB_F(C-\gamma_FC')/C^2$ and $C'=1/((1+\gamma_F)\ln 2)$, setting the derivative to zero gives $\xi\nu_c(C-\gamma_FC')=2\zeta C'$. Multiplying by $(1+\gamma_F)\ln 2$ turns $(C-\gamma_FC')(1+\gamma_F)\ln 2$ into $(1+\gamma_F)\ln(1+\gamma_F)-\gamma_F$. Monotonicity: the derivative of the left side is $\ln(1+\gamma_F)>0$. □

Interpretation: the left side is the marginal energy cost of shortening the floor by boosting its power; the right side is the marginal value of the released dimensions, both priced by the dual variables of the refinement problem.

Numerical illustration (synthetic). Table 16 uses a synthetic Gauss–Markov spectrum $\lambda_i=10^3(1+i/20)^{-2}$ with $k=2\times10^4$ coefficients, $a=0.9$, $\omega_i=(1-a^2)\lambda_i$, $W=5$ MHz, $T_g=8/30$ s, average signal-to-noise ratio 10 dB, a floor level $\theta_F=0.206$ set by $T_{\min}=2$ Mbps (credited 0.0482 bit per coefficient; payload $B_F/k=0.093$ with the firewall price), and the side-information recursion (9) iterated to convergence. Throughput is scaled to 720p at $G=8$. The numbers illustrate structure; they are not a performance claim.

The sweep is maximal at $\gamma_F=10.5$ dB, where the distortion is also minimal (5.399×10^{-3} per coefficient). Equation (32) predicts about 9.1 dB; at that point the throughput is 0.6% and the distortion 0.7% from the optimum. The 1.4 dB gap arises because (32) holds the side information fixed whereas, in the sweep, κ_i depends on the previous distortion through (9), and because the active set is discrete. The objective is flat near the optimum, so (32) is a reliable starting point and not an exact solution.

B.3 Robustness of the Open-Loop Gain Design

The transmitter measures its own residual energies ϑ_i exactly; what it cannot know are the receiver's side-information quality κ_i and the noise ν_i. Suppose the true values κ_i',ν_i' satisfy

$$\frac{\kappa_i}{1+\delta}\le\kappa_i'\le(1+\delta)\kappa_i,\qquad\frac{\nu_i}{1+\delta}\le\nu_i'\le(1+\delta)\nu_i,\qquad\kappa_i'\le\vartheta_i. \tag{33}$$

Table 17: Regret $\sum D'(x^*)/\sum D'(x')$ under (33): worst value over 300 random draws (random instances with $m = 60$ components), worst value for adversarial corners (parameters worse by $1+\delta$ on the design-active components, better elsewhere), and the bound of Proposition 15.

δ	Random draws	Corner	Bound $(1+\delta)^2$
0.10	1.0018	1.0002	1.2100
0.25	1.0064	1.0019	1.5625
0.50	1.0238	1.0073	2.2500
1.00	1.0622	1.0245	4.0000

Proposition 15 (Multiplicative regret)**.** *Let x^* be optimal for the nominal parameters, and x' optimal for the true ones with the same power budget and dimension limit. Under* (33) *and a receiver that uses the true κ_i', ν_i',*

$$\sum_i D_i'(x^*) \;\le\; (1+\delta)^2 \sum_i D_i'(x'). \tag{34}$$

Proof. For every gain x_i and active set, $1/\kappa' + x/\nu' \in \big[(1/\kappa + x/\nu)/(1+\delta),\, (1+\delta)(1/\kappa + x/\nu)\big]$, hence $D_i/(1+\delta) \le D_i' \le (1+\delta)D_i$. The constraints do not involve κ or ν, so x' is feasible nominally. Then $\sum D'(x^*) \le (1+\delta)\sum D(x^*) \le (1+\delta)\sum D(x') \le (1+\delta)^2 \sum D'(x')$. □

Corollary 8 (Scene cut)**.** *If the side information becomes useless, then $\kappa_i' = \vartheta_i$ and $\sum_i D_i'(x^*) \le \varrho_{\max} \sum_i D_i(x^*)$ with $\varrho_{\max} = \max_i \vartheta_i/\kappa_i$, the redundancy ratio of Section 5.*

Proof. With no side information the posterior variance is $(1/\lambda_i + [1/\theta_F - 1/\lambda_i]^+)^{-1} = \min\{\lambda_i, \theta_F\} = \vartheta_i$, so $\kappa_i'/\kappa_i \le \varrho_i$ and the same harmonic argument applies. □

Table 17 shows the bound is loose by a wide margin, as expected of a second-order effect around an optimum. In a scene-cut instance ($m = 60$, $\varrho_{\max} = 15.9$) the design gains give a distortion 3.02 times the nominal value, against the bound 15.9, and 1.23 times the distortion of the re-optimised gains.

Corollary 9 (The guarantee survives mismatch)**.** *Mismatch in κ_i, ν_i or the side information changes only the distortion above the floor. By Theorem 2, $\mathbb{P}[\mathcal{T}_{\text{eq}} < T_{\min}] \le \epsilon_F$ holds for every value of the mismatch.*

B.4 Open Items

The decoder is assumed to use the true κ_i' and ν_i'; the cost of estimating them (channel estimation error, pilot overhead) and the control loop that sets γ_F and the split from a delayed capacity estimate are the subject of the next section(s).

C Channel Estimation, Delayed Feedback, and Closed-Loop Control

C.1 Cost of Channel Estimation

Within a fading block of m channel uses the receiver estimates the gain from N_p pilot symbols of the same power as the data. For $h \sim \mathcal{CN}(0,1)$ and mean signal-to-noise ratio $\bar\gamma$ the minimum-mean-squared-error estimate $\hat h$ has error variance $\sigma_e^2 = 1/(1+N_p\bar\gamma)$, and $\hat h \sim \mathcal{CN}(0, 1-\sigma_e^2)$. Treating the estimation error as additional noise gives the effective signal-to-interference-and-noise ratio $\bar\gamma|\hat h|^2/(\sigma_e^2\bar\gamma + 1)$, and the rate

$$R(N_p) = \frac{m - N_p}{m}\, \mathbb{E}\log_2\Big(1 + \frac{\bar\gamma|\hat h|^2}{\sigma_e^2\bar\gamma + 1}\Big) \tag{35}$$

Table 18: Optimal number of pilots N_p^* (overhead in percent in parentheses) and the rate (35) in bit per channel use, for block length m and mean signal-to-noise ratio $\bar{\gamma}$; the last row gives the rate with perfect channel knowledge and no pilots. Rayleigh fading, 4×10^5 Monte-Carlo draws.

m	5 dB	10 dB	20 dB
50	5 (10.0), 1.361	4 (8.0), 2.421	3 (6.0), 5.155
100	8 (8.0), 1.455	6 (6.0), 2.552	4 (4.0), 5.351
200	11 (5.5), 1.527	9 (4.5), 2.650	6 (3.0), 5.499
500	18 (3.6), 1.593	14 (2.8), 2.741	10 (2.0), 5.636
perfect, no pilots	1.716	2.906	5.884

is an achievable lower bound [**?**]. Table 18 gives the maximiser N_p^* and the corresponding overhead $\varphi = N_p^*/m$ of Section 3.

Empirically the overhead falls roughly as $m^{-1/2}$ over this range; this is an observation, not a derived law. For the floor, γ_ℓ is replaced by the effective ratio above, and the information-density sampler of Section A must be extended accordingly; we do not do so here.

For the refinement layer the estimation error acts differently: its noise is proportional to the signal energy.

Proposition 16 (Allocation under channel-estimation error)**.** *Let component i with energy $u_i = \vartheta_i x_i$ see the noise $\nu_i + \sigma_i^2 u_i$, where σ_i^2 is the normalised estimation-error variance after equalisation, so that $D_i = \big(1/\kappa_i + u_i/(\vartheta_i(\nu_i + \sigma_i^2 u_i))\big)^{-1}$. Minimising $\sum_i D_i$ subject to $\sum_i u_i \le \mathcal{P}$ is a convex programme whose solution, for a multiplier $\xi > 0$, is*

$$u_i = \Big[\frac{1/z_i - \nu_i}{\sigma_i^2}\Big]^+, \qquad z_i = \frac{\sqrt{\xi}\, c_i}{\sqrt{\nu_i/\vartheta_i} + \sqrt{\xi}\, \nu_i/(\sigma_i^2 \vartheta_i)}, \qquad c_i = \frac{1}{\kappa_i} + \frac{1}{\sigma_i^2 \vartheta_i}, \tag{36}$$

with ξ the unique root of $\sum_i u_i(\xi) = \mathcal{P}$. Moreover, for every energy,

$$D_i \;\ge\; \Big(\frac{1}{\kappa_i} + \frac{1}{\sigma_i^2 \vartheta_i}\Big)^{-1}, \tag{37}$$

so the effective precision of a component cannot exceed $1/(\sigma_i^2 \vartheta_i)$. As $\sigma_i^2 \to 0$ the solution tends to that of Proposition 8.

Proof. Write $q(u) = u/(\vartheta(\nu + \sigma^2 u))$, increasing and concave, and $D = (1/\kappa + q)^{-1}$, convex and decreasing in q; the composition is convex, and the constraint is linear, so the Karush–Kuhn–Tucker conditions are sufficient. For $u > 0$ stationarity reads $D^2 q'(u) = \xi$ with $q'(u) = \nu/(\vartheta(\nu + \sigma^2 u)^2)$. With $z = 1/(\nu + \sigma^2 u)$ one has $1/D = c - \nu z/(\sigma^2 \vartheta)$ and $q' = \nu z^2/\vartheta$, so $z\sqrt{\nu/\vartheta} = \sqrt{\xi}\,(c - \nu z/(\sigma^2\vartheta))$, which solves to (36); $u \ge 0$ gives the positive part. The ceiling follows from $q(u) \uparrow 1/(\sigma^2\vartheta)$ as $u \to \infty$. The map $\xi \mapsto z_i$ is increasing, hence u_i is decreasing in ξ and the root is unique. □

At $N_p = 4$ pilots and $\bar{\gamma} = 10$ dB, $\sigma_e^2 = 1/41$, so no component can exceed an effective signal-to-noise ratio of $10 \log_{10} 41 = 16.1$ dB. Equation (36) was checked against a constrained numerical optimiser on ten random instances (maximum relative gap 4×10^{-14}) and against Proposition 8 at $\sigma^2 = 10^{-6}$ (total distortion 3.025342 versus 3.025335).

C.2 Delayed, Noisy Capacity Feedback

The floor design needs the mean signal-to-noise ratio of the coming group; the transmitter holds only a delayed, noisy report.

Table 19: Margin $z_\eta\sqrt{P_{\mathrm{pred}}(d)}$ and the resulting increase of the floor channel uses at nominal mean signal-to-noise ratio 15 and 20 dB. Illustrative parameters: $T_g = 8/30$ s, large-scale correlation time 2 s ($\rho = 0.875$), $\sigma_s = 3$ dB, $\sigma_m = 1$ dB, $\eta = 10^{-4}$ ($z_\eta = 3.72$).

Delay d	$\sqrt{P_{\mathrm{pred}}}$ [dB]	Margin [dB]	$+n_F$ at 15 dB [%]	$+n_F$ at 20 dB [%]
0	0.853	3.17	25.0	18.5
1	1.632	6.07	60.1	42.1
2	2.036	7.57	85.7	58.2
4	2.480	9.22	122.8	80.2
8	2.832	10.53	161.2	101.8
none	3.000	11.16	183.2	113.8

A7 The mean signal-to-noise ratio s_g of group g, in decibel, is a stationary Gauss–Markov process, $s_g - \mu_s = \rho(s_{g-1} - \mu_s) + w_g$, with stationary standard deviation σ_s and innovation variance $\sigma_s^2(1-\rho^2)$. The transmitter holds the report $\hat{s}_{g-d} = s_{g-d} + \varepsilon_{g-d}$ with $\varepsilon \sim \mathcal{N}(0, \sigma_m^2)$ independent, delivered after a delay of d groups. Fast fading within a group is carried by the diversity of the floor block and is not part of s_g.

A8 For a floor block sized for the design mean signal-to-noise ratio γ_d, the certified error bound of Theorem 1 is non-increasing in the actual mean signal-to-noise ratio.

A8 is not proved. For the certified $L = 16$ design of Table 9 (sized for 10 dB, $n_F = 297{,}376$) the dependence-testing estimate was 2.1×10^{-2}, 3.2×10^{-3}, 3.7×10^{-4}, 4×10^{-5} and 0 at $8, 9, 10, 11, 12$ dB (6×10^5 samples each).

Lemma 8 (Prediction variance)**.** *Under A7 the conditional law of s_g given all reports up to $g - d$ is Gaussian with variance*

$$P_{\mathrm{pred}}(d) = \rho^{2d} P_f + \sigma_s^2(1 - \rho^{2d}), \qquad P_f = D_\infty(1/\sigma_m^2) \text{ of (10) with } a = \rho,\ \omega = \sigma_s^2(1-\rho^2). \tag{38}$$

Proof. The Kalman filter for the scalar autoregressive state with measurement precision $\eta = 1/\sigma_m^2$ has the recursion of Lemma 3, whose steady state is Proposition 3; P_f is the posterior variance at the time of the latest report. Propagating d steps without measurements adds $\sigma_s^2(1-\rho^{2d})$ and scales the old variance by ρ^{2d}. □

Theorem 3 (Chance-constrained margin). Let $\hat{s}_{\mathrm{pred}}$ be the conditional mean, $z_\eta = Q^{-1}(\eta)$, and design the floor for $\gamma_d = \hat{s}_{\mathrm{pred}} - z_\eta\sqrt{P_{\mathrm{pred}}(d)}$ (decibel), certified with core failure probability ϵ_{core}. Under A7–A8 the floor fails with probability at most $\eta + \epsilon_{\mathrm{core}}$.

Proof. By Lemma 8, $\mathbb{P}[s_g < \gamma_d] = \eta$. On $\{s_g \geq \gamma_d\}$ the failure probability is at most ϵ_{core} by A8. □

The chance budget therefore splits as $\epsilon_F = \eta + \epsilon_{\mathrm{core}}$. The margin grows linearly in z_η whereas ϵ_{core} enters the floor size only through $Q^{-1}(\epsilon_{\mathrm{core}}/2)$, so most of the budget should be given to η.

Cost of a margin. To first order $n_F \propto 1/\log_2(1+\gamma)$, so a margin of M dB raises the floor by the factor $C(\gamma)/C(\gamma/10^{M/10})$, and the sensitivity is

$$\frac{\mathrm{d}\ln n_F}{\mathrm{d}\gamma_{\mathrm{dB}}}\Big|_{\text{per dB}} = -\frac{\gamma \ln 10/10}{(1+\gamma)\ln(1+\gamma)}, \tag{39}$$

that is 12.3%, 8.7%, 6.4% and 4.9% of the floor channel uses per decibel at $5, 10, 15, 20$ dB.

With these parameters a floor sized without feedback needs a margin of 11.2 dB and is infeasible at 10 dB nominal (the design point would be below 0 dB); with a one-group delay the

margin is 6.1 dB. Timely feedback is therefore a structural requirement of the architecture, and its value is quantified by Table 19. The parameters are illustrative; they must be measured for the target link.

C.3 Closed-Loop Control of Gains and Split

The refinement gains are recomputed every group from the current estimates; an error of e dB in the reported noise level is a mismatch $\delta = 10^{|e|/10} - 1$ in (33), so Proposition 15 bounds the loss by $(1+\delta)^2$ ($\delta = 0.26$ for 1 dB).

The floor power and split follow Proposition 14. Define $\Gamma(\gamma_F) = f^{-1}\big(2\zeta(\gamma_F)/(\xi(\gamma_F)\nu_c)\big)$ with $f(\gamma) = (1+\gamma)\ln(1+\gamma) - \gamma$, where ξ, ζ are the multipliers of the refinement problem solved at γ_F, and iterate with damping β:

$$\gamma_F^+ = (1-\beta)\,\gamma_F + \beta\,\Gamma(\gamma_F). \tag{40}$$

Proposition 17 (Local convergence of the controller)**.** *Let Γ be continuously differentiable near a fixed point γ^* with slope $s = \Gamma'(\gamma^*) < 0$. The iteration* (40) *converges locally if and only if $0 < \beta < 2/(1+|s|)$, with local rate $|1 - \beta(1+|s|)|$; $\beta = 1/(1+|s|)$ removes the first-order error.*

Proof. The linearisation of (40) at γ^* has slope $1 - \beta + \beta s = 1 - \beta(1+|s|)$; local convergence is equivalent to its modulus being below one. □

The slope is negative because f is increasing while $2\zeta/(\xi\nu_c)$ decreases in γ_F (Table 16): boosting the floor power lowers the marginal price of a dimension. On the synthetic model of Section B ($k = 8 \times 10^3$) the undamped iteration ($\beta = 1$) oscillates with successive-difference ratios between -0.59 and -0.51 and approaches 9.4 dB from both 6 and 14 dB; the slope is therefore $s \approx -0.52$. With $\beta = 0.66 \approx 1/(1+0.52)$ both starts reach 9.41 dB in two steps (throughput 9.658 Mbps against 9.65 Mbps for the undamped iteration after seven steps). The fixed point is the stationary point of Proposition 14, which Section B found about 1 dB below the exhaustive optimum.

Proposition 18 (Separation of safety and performance)**.** *Let $\mathcal{G}$ be a finite grid of floor designs (γ_F, n_F), each certified by the procedure of Proposition 11 at the core level ϵ_{core} for a given margin. If the controller* (40) *is restricted to $\mathcal{G}$ (rounding each iterate to the nearest certified design not below the requested floor power), the floor guarantee of Theorem 3 holds for every controller trajectory, whether or not the iteration converges.*

Proof. Theorem 3 depends only on the certified design in force at each group and on the margin rule, not on how the design was selected. □

Theorem 4 (System guarantee with delayed feedback)*.* Under A1–A8, with the margin rule of Theorem 3, a certified design in force at each group, and arbitrary (including divergent) controller trajectories and refinement-layer mismatch,

$$\mathbb{P}\big[\mathcal{T}_{\text{eq}}(g,h) < T_{\min}\big] \le \eta + \epsilon_{\text{core}} + \epsilon_q$$

for every group g, where ϵ_q is the quantiser violation probability of A6. The refinement layer affects only the throughput above $T_{\min}$.

Proof. Combine Theorem 2 (with the floor failure probability bounded by Theorem 3) and Proposition 7. □

C.4 Open Items

The evaluation protocol on real video (estimation of λ_i, ω_i, a_i, baselines, ablations) and the discussion of limitations follow. Two extensions remain outside this section: the floor certificate with channel estimation error and clipping included, and the validity of A7 and A8 on measured links.

Table 20: Additional verification checks.

Quantity	Result
Limiter gain and distortion (29) (4×10^6 draws, $\Lambda = 1, 2, 4, 8$)	$\lvert\Delta\alpha\rvert \le 3 \times 10^{-4}$, $\lvert\Delta\sigma_d^2/\sigma^2\rvert \le 2 \times 10^{-5}$
Mismatch regret Proposition 15 (300 random instances per δ)	observed worst 1.0018 to 1.0622 against bound 1.21 to 4 (Table 17)
Allocation under estimation error (36)	maximum relative gap to numerical optimum 4×10^{-14}
Prediction margin (38), $d = 1$	6.07 dB (Table 19)
Controller (40), $\beta = 0.66$	settles in two steps from 6 and 14 dB

D Evaluation Protocol and Pilot Results

D.1 Aims

The protocol tests five statements derived from the preceding sections:

H1 When the floor block is decoded, the reconstruction quality is at least that of the floor level (Proposition 7).

H2 The analytic distortion predicted by the model of Sections 4 and 5 matches the measured distortion.

H3 Receiver-side prediction lowers distortion, and the firewalled floor costs the payload π_g of Proposition 6.

H4 The hybrid design improves on a linear (SoftCast-type) design at the bandwidth ratios of interest.

H5 The design competes with a digital codec followed by an ideal channel code.

H5 is the strongest test; we report it as measured.

D.2 Material, Operating Points, and Scope of the Pilot

The intended material is uncompressed sequences, for example the raw sequences of the Xiph.org video test media collection. Such sequences were not available to us. The pilot uses one clip bundled with a public software package: a 1280×720, 25 frames-per-second animation, decoded from a 1.2 Mbps H.264 file, reduced to 640×360 by area averaging and cropped to 640×352 (so that the chroma planes divide by 8), 128 frames, 16 groups of $G = 8$. *The source is therefore previously compressed, animated, and of low entropy; it favours digital codecs, and the absolute values below are not representative of raw video.*

Operating points are defined by bits per sample so that they correspond to the 2 and 10 Mbps points at 720p: 2 Mbps at 720p25 is 0.0579 bit per sample and 10 Mbps is 0.289 bit per sample (Table 1).

D.3 Parameter Estimation and Metadata

For each group the encoder measures, from the source frames only, the chunk variance λ_j, the regression coefficient a_j of the chunk on the temporal-DC prediction from the previous *original* last frame, and the residual variance ω_j; the values are quantised to 1 dB (and a_j to 1/32) and sent as metadata: 12 bits per chunk plus 5 bits per temporal-DC chunk, 36,360 bits per group. The side information is $z = \sqrt{G}\,\mathrm{DCT}_2(\hat{\boldsymbol{x}}_{\mathrm{last}})$ placed in the temporal-DC band of the reconstruction

Table 21: Pilot operating point.

Quantity	Value
Frame format	640×352, 4:2 : 0, 25 frames per second
Group	$G = 8$ frames, $T_g = 0.32$ s, $k = 2{,}703{,}360$ coefficients
Credited floor (2 Mbps equivalent)	489 kbit/s, $B_c = 156{,}444$ bit per group
Ceiling (10 Mbps equivalent)	2.444 Mbit/s, the capacity at 10 dB
Channel	additive white Gaussian noise, $n_{\text{eff}} = 226{,}113$ complex uses per group (pilot overh
Chunks	2,880: temporal index $\times$ 15 frequency zones $(u + v)$ $\times$ regions (4×4 luma, 2×2

of the previous group (zero-motion prediction, a pessimistic stand-in for decoder-side motion compensation); the prior mean of band j is $a_j z$.

Lemma 9 (Temporal mixing of the side-information error)**.** *Let the side information of group g be built from the last frame of the previous reconstruction, and let w_t be the weight with which the temporal coefficient t enters the last frame (the last row of the inverse temporal transform, $\sum_t w_t^2 = 1$). If the errors of the temporal bands of group $g - 1$ in the same zone and region are uncorrelated with variances $D_{t,j}(g - 1)$, then the prediction error of the temporal-DC band satisfies*

$$\mu_{0,j}(g) = \omega_j + a_j^2 \, G \sum_{t=0}^{G-1} w_t^2 \, D_{t,j}(g - 1), \tag{41}$$

in place of (9)*; for $G = 1$ the two coincide.*

Proof. The error of the last frame in coefficient (u, v) is $\sum_t w_t \varepsilon_t$ and that of z is $\sqrt{G}$ times it, so its variance is $G \sum_t w_t^2 D_{t,j}$; adding the innovation variance gives (41). □

In the pilot, using (41) instead of the per-band recursion changed the predicted quality by up to 0.2 dB.

D.4 Baselines, Ablations, and Metrics

Baselines. (i) A hybrid codec (H.264, library `libx264` through ffmpeg 7.0.2, two-pass) at the bit rate $n_{\text{eff}} C(\gamma)/T_g$, that is, followed by an *ideal* capacity-achieving channel code with oracle rate adaptation, in two configurations: default (preset medium, B-frames, long groups) and latency-matched (zero-latency tuning, group length 8). (ii) A chunk-linear transmission without floor or side information, in the manner of SoftCast [16]; this is our implementation at chunk granularity, not the original system. A learned joint-coding baseline [3] was not evaluated.

Ablations. Without side information; without the floor; without both (the linear baseline). The no-firewall variant (a floor that uses the previous reconstruction) is evaluated by payload only.

Metrics. Peak signal-to-noise ratio computed jointly over the luma and both chroma planes, per group, reported as the mean over groups and the minimum over groups; the fraction n_F/n_{eff} of channel uses taken by the floor; the model-minus-measured gap; the floor payload relative to B_c.

D.5 Findings

H1 (supported, scope limited).

The floor-only reconstruction reaches 35.8 dB on average and never falls below 31.6 dB; every full reconstruction in Table 23 stays above its group's floor-only quality. This is

Table 22: Floor statistics, mean over 16 groups.

Quantity	Value
Fraction of coefficients coded by the floor	8.04%
Floor-only peak signal-to-noise ratio, mean (minimum)	35.78 (31.63) dB
Payload, Laplacian-model code length	260,627 bit
Payload, Gaussian information function $\mathcal{R}_g^{\text{intra}}$	298,837 bit (model-to-model ratio 0.885)
Metadata	36,360 bit
Floor block / credited payload B_c	1.90 (maximum over groups 3.13)
Lowest signal-to-noise ratio with a feasible floor in all groups	between 4 and 6 dB

Table 23: Peak signal-to-noise ratio in dB, mean over 16 groups (minimum over groups in parentheses), additive white Gaussian noise channel, signal-to-noise ratio γ. The digital rows are H.264 at the capacity-matched rate $n_{\text{eff}}C(\gamma)/T_g$ (968, 1281, 1637, 2028, 2444, 2879 kbit/s) with an ideal code. "infeasible" means the floor block does not fit in the group for at least one group. Single noise realisation.

γ [dB]	2	4	6	8	10	12
Floor + side information (proposed)	infeasible	infeasible	38.43 (34.79)	39.52 (35.60)	40.56 (36.27)	41.42 (36.84)
model prediction	—	—	38.78	39.88	40.81	41.60
floor share n_F/n_{eff}	—	—	0.569	0.460	0.381	0.324
Floor, no side information	infeasible	infeasible	38.25 (34.79)	39.22 (35.59)	40.11 (36.27)	40.92 (36.83)
No floor, side information	27.56 (25.01)	28.99 (26.72)	30.44 (28.44)	31.94 (30.09)	33.48 (31.49)	35.00 (32.86)
Linear, no floor, no side information	24.85 (24.35)	26.52 (25.92)	28.19 (27.49)	29.89 (29.07)	31.61 (30.67)	33.33 (32.22)
H.264, default preset	43.16 (41.82)	44.34 (42.86)	45.40 (43.80)	46.36 (44.61)	47.23 (45.38)	48.05 (46.11)
H.264, zero-latency	38.87 (37.80)	40.40 (39.27)	41.73 (40.47)	42.90 (41.52)	43.95 (42.58)	44.92 (43.55)

a statement conditional on the floor block being decoded; floor outage under fading was not simulated.

H2 (supported).
Model and measurement differ by 0.35, 0.36, 0.26 and 0.18 dB at $6, 8, 10, 12$ dB for the proposed design, by at most 0.05 dB without side information, and by 0.59 to 0.78 dB without the floor.

H3 (partly supported).
Side information adds 0.18 to 0.50 dB (growing with signal-to-noise ratio) with the floor and 1.67 to 2.71 dB without it; the zero-motion predictor is weak. The firewall price is visible as the Gaussian-versus-operational payload gap of Table 22, but the payload of a floor that used the previous reconstruction was not measured.

H4 (supported).
At the bandwidth ratio $2n_{\text{eff}}/k \approx 0.17$ the hybrid design exceeds the linear design by 10.2, 9.6, 8.9 and 8.1 dB at $6, 8, 10, 12$ dB.

H5 (not supported on this source).
The proposed design is below zero-latency H.264 with an ideal code by 3.3 to 3.5 dB at every signal-to-noise ratio and below the default preset by about 7 dB; the floor block costs 1.90 times the credited payload, and per bit the floor-only reconstruction is about 3 dB below zero-latency H.264. The digital baselines here receive oracle rate adaptation and an ideal code, and the source is pre-compressed, which favours them.

The measured costs have identifiable sources: the firewall price (Table 5), metadata (12% of the floor block), scalar quantisation with a static zeroth-order code length against context-adaptive entropy coding and motion compensation in the codec, and a floor that does not exploit side information. None of these was tuned.

Table 24: Further pilot comparisons. Margin experiment: the floor and metadata are sized for $\gamma - 3$ dB while refinement uses the actual γ. Per-bit comparison: peak signal-to-noise ratio of the floor-only reconstruction against H.264 at comparable rates.

Experiment	Result
Margin 3 dB, proposed, $\gamma = 8$ dB	infeasible
Margin 3 dB, proposed, $\gamma = 10$ dB	39.46 (35.72)
Margin 3 dB, proposed, $\gamma = 12$ dB	40.55 (36.41)
Floor-only, 928 kbit/s block (297,000 bit per 0.32 s)	35.78 (31.63)
H.264 zero-latency, 489 kbit/s	35.28 (34.24)
H.264 zero-latency, 968 kbit/s	38.87 (37.80)
H.264 default, 489 kbit/s	40.40 (39.33)

D.6 Limitations of the Pilot

One clip, previously compressed and animated; 16 groups; additive white Gaussian noise only, with the floor sized at the true signal-to-noise ratio (oracle feedback) and no fading, so floor outage and the digital cliff are not exercised; one noise realisation; encoder statistics computed per group from the source (oracle estimates, with their metadata cost counted); a zero-motion side-information predictor; learned and rateless baselines absent; encoder complexity not measured. The comparison therefore bounds the proposed design from below in peak signal-to-noise ratio under oracle conditions and says nothing about the regime where a fixed-rate digital chain fails.

D.7 Open Items

Repeat the protocol on raw sequences with fading, delayed feedback (Table 19) and without oracle rate adaptation; add decoder-side motion compensation; reduce the metadata (a parametric variance field); replace the static code-length model by a context-adaptive one; and add an analogue fallback below the floor-feasibility threshold. Discussion, limitations and conclusion follow.

E Reproduction of the Pilot

Files: `evallab.py` (chunk layout, statistics, floor design, allocation, decoder), `runeval.py` (modes `grid`, `margin`, `floor`), `runh264.py`. Data preparation (clip bundled with `scikit-video`, `bigbuckbunny.mp4`):

```
ffmpeg -i bigbuckbunny.mp4 -vf "scale=640:360:flags=area,crop=640:352:0:4" \
       -frames:v 128 -pix_fmt yuv420p -f rawvideo bbb352.yuv
python runeval.py grid; python runeval.py margin; python runeval.py floor
python runh264.py
```

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